

cleaning

April 2, 2026

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[11]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set global font settings - consistent across all plots
plt.rcParams['font.family'] = 'Times New Roman'
plt.rcParams['axes.labelsize'] = 12
plt.rcParams['xtick.labelsize'] = 10
plt.rcParams['ytick.labelsize'] = 10
plt.rcParams['axes.titlesize'] = 12
plt.rcParams['legend.fontsize'] = 10
plt.rcParams['figure.titlesize'] = 12
plt.rcParams['figure.dpi'] = 100

# Define consistent styling function for all plots with grayscale
def style_plot(ax, title, xlabel, ylabel, legend=True, grid=True):
    """Apply consistent styling to plots with grayscale theme"""
    ax.set_title(title, fontsize=12, fontweight='bold', pad=15, color='black')
    ax.set_xlabel(xlabel, fontsize=12, labelpad=10, color='black')
    ax.set_ylabel(ylabel, fontsize=12, labelpad=10, color='black')
    if grid:
        # Only show horizontal grid lines
        ax.grid(True, axis='y', alpha=0.3, linestyle='-', linewidth=0.5,
        color='#666666')
        # Disable vertical grid lines
        ax.grid(False, axis='x')
    if legend:
        ax.legend(fontsize=10, frameon=True, loc='upper left',
        bbox_to_anchor=(1, 1))
    ax.tick_params(axis='both', which='major', labelsize=10, colors='black')
    ax.set_facecolor('#f5f5f5')
    ax.spines['top'].set_color('#cccccc')
    ax.spines['bottom'].set_color('#cccccc')
    ax.spines['left'].set_color('#cccccc')
    ax.spines['right'].set_color('#cccccc')
    return ax
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# Load dataset
data = pd.read_csv("datasets/family_anxiety_14_dataset.csv") # training dataset
# data = pd.read_csv("datasets/enhanced_anxiety_dataset.csv") # testing dataset

# Display basic dataset information
print("DATASET DESCRIPTION")
print(f"Dataset shape: {data.shape[0]} rows, {data.shape[1]} columns")
print(f"Column names: {list(data.columns)}")
print(f>Data types:\n{data.dtypes}")
print(f"Missing values:\n{data.isnull().sum()}")

# Convert Yes/No columns to binary using map() instead of replace to avoid
↳warning
yes_no_cols = ['Smoking', 'Family History of Anxiety', 'Dizziness',
               'Medication', 'Recent Major Life Event']
for col in yes_no_cols:
    data[col] = data[col].map({'Yes': 1, 'No': 0})

# Creating target column: Diagnosed if Anxiety Level > 4
data['Diagnosed'] = (data['Anxiety Level (1-10)'] > 4).astype(int)

print("NEW TARGET COLUMN DIAGNOSED")
print(data[['Anxiety Level (1-10)', 'Diagnosed']].head())
print(f"Diagnosed distribution: 0 = {sum(data['Diagnosed']==0)}, 1 =
↳{sum(data['Diagnosed']==1)}")

# Features vs target
features = data.drop(columns=['Diagnosed'])
target = data['Diagnosed']

# Display min and max for numeric features after conversion
numeric_features = features.select_dtypes(include=[np.number])

print("NUMERIC FEATURES MIN AND MAX VALUES")
for col in numeric_features.columns:
    print(f"{col}: Min = {numeric_features[col].min()}, Max =
↳{numeric_features[col].max()}")

# statistical summary
print("STATISTICAL SUMMARY OF NUMERIC FEATURES")
print(numeric_features.describe())

# OUTLIER DETECTION FOR CONTINUOUS COLUMNS
print("OUTLIER DETECTION FOR CONTINUOUS COLUMNS")

# Identify continuous columns

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continuous_cols = ['Age', 'Anxiety Level (1-10)', 'Sleep Hours', 'Physical_
↳Activity (hrs/week)']

print(f"Continuous columns: {continuous_cols}")

# IQR Method for outlier detection
def detect_outliers_iqr(data, columns):
    outlier_summary = {}

    for col in columns:
        if col in data.columns:
            Q1 = data[col].quantile(0.25)
            Q3 = data[col].quantile(0.75)
            IQR = Q3 - Q1
            lower_bound = Q1 - 1.5 * IQR
            upper_bound = Q3 + 1.5 * IQR

            outliers = data[(data[col] < lower_bound) | (data[col] >_
↳upper_bound)]
            outlier_indices = outliers.index.tolist()

            outlier_summary[col] = {
                'Q1': Q1,
                'Q3': Q3,
                'IQR': IQR,
                'lower_bound': lower_bound,
                'upper_bound': upper_bound,
                'outlier_count': len(outliers),
                'outlier_percentage': (len(outliers) / len(data)) * 100,
                'outlier_indices': outlier_indices,
                'outlier_values': outliers[col].tolist()
            }

    return outlier_summary

# Z-Score Method for outlier detection
def detect_outliers_zscore(data, columns, threshold=3):
    outlier_summary = {}

    for col in columns:
        if col in data.columns:
            mean = data[col].mean()
            std = data[col].std()
            z_scores = np.abs((data[col] - mean) / std)

            outliers = data[z_scores > threshold]
            outlier_indices = outliers.index.tolist()

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        outlier_summary[col] = {
            'mean': mean,
            'std': std,
            'threshold': threshold,
            'outlier_count': len(outliers),
            'outlier_percentage': (len(outliers) / len(data)) * 100,
            'outlier_indices': outlier_indices,
            'outlier_values': outliers[col].tolist(),
            'max_zscore': z_scores.max()
        }

    return outlier_summary

# Detect outliers using both methods
iqr_outliers = detect_outliers_iqr(data, continuous_cols)
zscore_outliers = detect_outliers_zscore(data, continuous_cols)

# Display IQR method results
print("IQR METHOD OUTLIER RESULTS")
for col, stats in iqr_outliers.items():
    print(f"{col}: {stats['outlier_count']} outliers, {stats['outlier_percentage']:.2f}%")
    print(f"    Bounds: [{stats['lower_bound']:.2f}, {stats['upper_bound']:.2f}]")
    if stats['outlier_count'] > 0:
        print(f"    Outlier values: {stats['outlier_values']}")

# Display Z-Score method results
print("ZSCORE METHOD OUTLIER RESULTS")
for col, stats in zscore_outliers.items():
    print(f"{col}: {stats['outlier_count']} outliers, {stats['outlier_percentage']:.2f}%")
    print(f"    Max ZScore: {stats['max_zscore']:.2f}")
    if stats['outlier_count'] > 0:
        print(f"    Outlier values: {stats['outlier_values']}")

# Numeric features correlation heatmap showing only high correlations
print("HIGH CORRELATIONS ABS > 0.2")
corr_matrix = numeric_features.corr(method='spearman')

# Create a mask for correlations with absolute value > 0.2
mask = np.abs(corr_matrix) > 0.2

# Remove diagonal
np.fill_diagonal(mask.values, False)

# Get pairs with high correlation

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high_corr_pairs = []
for i in range(len(corr_matrix.columns)):
    for j in range(i+1, len(corr_matrix.columns)):
        if mask.iloc[i, j]:
            high_corr_pairs.append({
                'Feature 1': corr_matrix.columns[i],
                'Feature 2': corr_matrix.columns[j],
                'Correlation': corr_matrix.iloc[i, j]
            })

# Display high correlation pairs
if high_corr_pairs:
    high_corr_df = pd.DataFrame(high_corr_pairs)
    high_corr_df = high_corr_df.sort_values('Correlation', ascending=False)
    print(high_corr_df.to_string(index=False))
else:
    print("No correlations with absolute value > 0.2 found")

# Create heatmap only for high correlations with grayscale styling - UNIFORM
FIGURE SIZE (8,6)
if high_corr_pairs:
    fig, ax = plt.subplots(figsize=(8, 6))

    high_corr_features = list(set([item['Feature 1'] for item in
    high_corr_pairs] +
                                [item['Feature 2'] for item in
    high_corr_pairs]))

    high_corr_matrix = corr_matrix.loc[high_corr_features, high_corr_features]

    # Use grayscale colormap for correlation heatmap
    sns.heatmap(high_corr_matrix, annot=True, fmt=".2f",
                cmap="Greys", center=0, linewidths=0.2, square=True,
                annot_kws={'size': 10, 'color': 'black'},
                cbar_kws={'label': 'Correlation Coefficient', 'shrink': 0.8},
    ax=ax)

    ax = style_plot(ax, "Feature Correlation Matrix ( $|r| > 0.2$ )",
                    "Features", "Features", legend=False, grid=False)
    plt.tight_layout()
    plt.show()

# 1. Target distribution plot - UNIFORM FIGURE SIZE (8,6)
fig, ax = plt.subplots(figsize=(8, 6))
bars = sns.countplot(x=data['Diagnosed'], palette=['#a0a0a0', '#404040'],
                    legend=False, ax=ax, edgecolor='black', linewidth=1)
ax = style_plot(ax, "Target Variable Distribution",
                "Diagnosis Status (0 = Non-Diagnosed, 1 = Diagnosed)",

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        "Count", legend=False)

total = len(data)
for bar in bars.patches:
    count = int(bar.get_height())
    percentage = count/total*100
    ax.text(bar.get_x() + bar.get_width()/2., bar.get_height() + 5,
            f'{count}\n({percentage:.1f}%)', ha='center', va='bottom',
            fontsize=10, fontweight='bold', color='black')

plt.tight_layout()
plt.show()

# 2. Feature distribution - UNIFORM FIGURE SIZE (8,6)
numeric_cols = numeric_features.columns.tolist()

for col in numeric_cols:
    fig, ax = plt.subplots(figsize=(8, 6))

    diagnosed_data = data[data['Diagnosed'] == 1][col]
    nondiagnosed_data = data[data['Diagnosed'] == 0][col]

    # Use grayscale colors
    ax.hist(nondiagnosed_data, bins=15, alpha=0.6, color='#a0a0a0',
            label=f'Non-Diagnosed (n={len(nondiagnosed_data)})',
            edgecolor='black', linewidth=0.5)
    ax.hist(diagnosed_data, bins=15, alpha=0.8, color='#404040',
            label=f'Diagnosed (n={len(diagnosed_data)})',
            edgecolor='black', linewidth=0.5)

    ax.axvline(nondiagnosed_data.mean(), color='#606060', linestyle='--',
            linewidth=2, label=f'Mean Non-Diagnosed: {nondiagnosed_data.
↵mean():.2f}')
    ax.axvline(diagnosed_data.mean(), color='black', linestyle='--',
            linewidth=2, label=f'Mean Diagnosed: {diagnosed_data.mean():.
↵2f}')

    ax = style_plot(ax, f'{col} Distribution by Diagnosis Status', col,
↵'Frequency')
    plt.tight_layout()
    plt.show()

# 3. Density plots - UNIFORM FIGURE SIZE (8,6)
for col in numeric_cols:
    fig, ax = plt.subplots(figsize=(8, 6))

    # Use grayscale colors

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subset0 = data[data['Diagnosed'] == 0]
subset1 = data[data['Diagnosed'] == 1]

sns.kdeplot(data=subset0, x=col, color='#808080',
            label=f'Non-Diagnosed (n={len(subset0)})',
            fill=True, alpha=0.4, linewidth=2, ax=ax)
sns.kdeplot(data=subset1, x=col, color='#404040',
            label=f'Diagnosed (n={len(subset1)})',
            fill=True, alpha=0.6, linewidth=2, ax=ax)

ax = style_plot(ax, f'{col} Density by Diagnosis Status', col, 'Density')
plt.tight_layout()
plt.show()

# Skewness and kurtosis for numeric features
skew_kurt = pd.DataFrame({
    'Skewness': numeric_features.skew(),
    'Kurtosis': numeric_features.kurtosis()
}).sort_values(by='Skewness', key=abs, ascending=False)

print("SKEWNESS AND KURTOSIS OF FEATURES")
print(skew_kurt)

# Categorical features summary
categorical_features = features.select_dtypes(include=['object', 'category']).
    columns

print("CATEGORICAL FEATURES SUMMARY")
for col in categorical_features:
    print(f"Categorical Feature: {col}")
    print(f"Number of unique values: {data[col].nunique()}")
    print("Value counts top 10:")
    print(data[col].value_counts().head(10))

# 4. Paired bar chart for binary features - UNIFORM FIGURE SIZE (8,6)
print("\n" + "="*60)
print("BINARY FEATURES PERCENTAGES BY DIAGNOSIS GROUP")

binary_cols = ['Smoking', 'Family History of Anxiety', 'Dizziness',
               'Medication', 'Recent Major Life Event']
feature_display_names = ['Smoking', 'Family History', 'Dizziness',
                        'Medication', 'Recent Life Event']

non_diagnosed_pcts = []
diagnosed_pcts = []

for col in binary_cols:

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diag_1_pct = data[data['Diagnosed'] == 1][col].mean() * 100
diag_0_pct = data[data['Diagnosed'] == 0][col].mean() * 100
non_diagnosed_pcts.append(diag_0_pct)
diagnosed_pcts.append(diag_1_pct)

# UNIFORM FIGURE SIZE (8,6) to match distribution graphs
fig, ax = plt.subplots(figsize=(8, 6))

x = np.arange(len(feature_display_names))
width = 0.35

# Use grayscale colors with consistent styling
bars1 = ax.bar(x - width/2, non_diagnosed_pcts, width,
               label='Non-Diagnosed', color='#a0a0a0',
               edgecolor='black', linewidth=1)
bars2 = ax.bar(x + width/2, diagnosed_pcts, width,
               label='Diagnosed', color='#404040',
               edgecolor='black', linewidth=1)

# Set xticks with consistent font size
ax.set_xticks(x)
ax.set_xticklabels(feature_display_names, fontsize=10)
ax.set_ylim(0, 100)
ax.set_yticks(np.arange(0, 101, 20))
ax.set_yticklabels([f'{i}%' for i in np.arange(0, 101, 20)], fontsize=10)

# Apply consistent styling using the same style_plot function
ax = style_plot(ax, "Binary Clinical Indicators: Non-Diagnosed vs Diagnosed",
               "Clinical Indicators", "Percentage")

# Add value labels with consistent font size
for bars in [bars1, bars2]:
    for bar in bars:
        height = bar.get_height()
        ax.text(bar.get_x() + bar.get_width()/2., height + 1,
                f'{height:.1f}%', ha='center', va='bottom',
                fontsize=10, fontweight='bold', color='black')

plt.tight_layout()
plt.show()

print("\nBinary Features - Percentage by Diagnosis Group:")
print("-" * 60)
for i, feature in enumerate(feature_display_names):
    print(f"{feature:20} Non-Diagnosed: {non_diagnosed_pcts[i]:.1f}% |  □
↳Diagnosed: {diagnosed_pcts[i]:.1f}% | Difference: {diagnosed_pcts[i] -
↳non_diagnosed_pcts[i]:.1f}%")

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# Save cleaned dataset
data.to_csv("training_dataset.csv", index=False)
# data.to_csv("testing_dataset.csv", index=False)
print("\n" + "="*60)
print("Dataset saved")
print(f"Final dataset shape: {data.shape[0]} rows, {data.shape[1]} columns")

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DATASET DESCRIPTION

Dataset shape: 11000 rows, 19 columns

Column names: ['Age', 'Gender', 'Occupation', 'Sleep Hours', 'Physical Activity (hrs/week)', 'Caffeine Intake (mg/day)', 'Alcohol Consumption (drinks/week)', 'Smoking', 'Family History of Anxiety', 'Stress Level (1-10)', 'Heart Rate (bpm)', 'Breathing Rate (breaths/min)', 'Sweating Level (1-5)', 'Dizziness', 'Medication', 'Therapy Sessions (per month)', 'Recent Major Life Event', 'Diet Quality (1-10)', 'Anxiety Level (1-10)']

Data types:

Age	int64
Gender	object
Occupation	object
Sleep Hours	float64
Physical Activity (hrs/week)	float64
Caffeine Intake (mg/day)	int64
Alcohol Consumption (drinks/week)	int64
Smoking	object
Family History of Anxiety	object
Stress Level (1-10)	int64
Heart Rate (bpm)	int64
Breathing Rate (breaths/min)	int64
Sweating Level (1-5)	int64
Dizziness	object
Medication	object
Therapy Sessions (per month)	int64
Recent Major Life Event	object
Diet Quality (1-10)	int64
Anxiety Level (1-10)	float64

dtype: object

Missing values:

Age	0
Gender	0
Occupation	0
Sleep Hours	0
Physical Activity (hrs/week)	0
Caffeine Intake (mg/day)	0
Alcohol Consumption (drinks/week)	0
Smoking	0
Family History of Anxiety	0
Stress Level (1-10)	0

Heart Rate (bpm)	0
Breathing Rate (breaths/min)	0
Sweating Level (1-5)	0
Dizziness	0
Medication	0
Therapy Sessions (per month)	0
Recent Major Life Event	0
Diet Quality (1-10)	0
Anxiety Level (1-10)	0

dtype: int64

NEW TARGET COLUMN DIAGNOSED

Anxiety Level (1-10)	Diagnosed
0	3.0 0
1	4.0 0
2	6.0 1
3	2.0 0
4	4.0 0

Diagnosed distribution: 0 = 7075, 1 = 3925

NUMERIC FEATURES MIN AND MAX VALUES

Age: Min = 18, Max = 64

Sleep Hours: Min = 0.2, Max = 12.5

Physical Activity (hrs/week): Min = 0.0, Max = 11.2

Caffeine Intake (mg/day): Min = 0, Max = 599

Alcohol Consumption (drinks/week): Min = 0, Max = 19

Smoking: Min = 0, Max = 1

Family History of Anxiety: Min = 0, Max = 1

Stress Level (1-10): Min = 1, Max = 10

Heart Rate (bpm): Min = 60, Max = 119

Breathing Rate (breaths/min): Min = 12, Max = 29

Sweating Level (1-5): Min = 1, Max = 5

Dizziness: Min = 0, Max = 1

Medication: Min = 0, Max = 1

Therapy Sessions (per month): Min = 0, Max = 9

Recent Major Life Event: Min = 0, Max = 1

Diet Quality (1-10): Min = 1, Max = 10

Anxiety Level (1-10): Min = 1.0, Max = 10.0

STATISTICAL SUMMARY OF NUMERIC FEATURES

	Age	Sleep Hours	Physical Activity (hrs/week) \
count	11000.000000	11000.000000	11000.000000
mean	40.414818	6.772427	2.916464
std	13.349472	1.615210	1.836844
min	18.000000	0.200000	0.000000
25%	29.000000	5.600000	1.400000
50%	40.000000	6.800000	2.800000
75%	52.000000	7.900000	4.200000
max	64.000000	12.500000	11.200000

Caffeine Intake (mg/day) Alcohol Consumption (drinks/week) \

count	11000.000000	11000.000000
mean	313.291364	9.725545
std	172.920481	5.716079
min	0.000000	0.000000
25%	163.000000	5.000000
50%	326.000000	10.000000
75%	462.000000	15.000000
max	599.000000	19.000000

	Smoking	Family History of Anxiety	Stress Level (1-10) \
count	11000.000000	11000.000000	11000.000000
mean	0.226273	0.133545	5.808545
std	0.418437	0.340179	2.941883
min	0.000000	0.000000	1.000000
25%	0.000000	0.000000	3.000000
50%	0.000000	0.000000	6.000000
75%	0.000000	0.000000	8.000000
max	1.000000	1.000000	10.000000

	Heart Rate (bpm)	Breathing Rate (breaths/min)	Sweating Level (1-5) \
count	11000.000000	11000.000000	11000.000000
mean	90.994000	20.790909	3.081273
std	17.138795	5.181014	1.401405
min	60.000000	12.000000	1.000000
25%	76.000000	16.000000	2.000000
50%	92.000000	21.000000	3.000000
75%	106.000000	25.000000	4.000000
max	119.000000	29.000000	5.000000

	Dizziness	Medication	Therapy Sessions (per month) \
count	11000.000000	11000.000000	11000.000000
mean	0.517909	0.508818	4.684727
std	0.499702	0.499945	2.852922
min	0.000000	0.000000	0.000000
25%	0.000000	0.000000	2.000000
50%	1.000000	1.000000	5.000000
75%	1.000000	1.000000	7.000000
max	1.000000	1.000000	9.000000

	Recent Major Life Event	Diet Quality (1-10)	Anxiety Level (1-10)
count	11000.000000	11000.000000	11000.000000
mean	0.522818	5.212727	4.149455
std	0.499502	2.879370	2.115930
min	0.000000	1.000000	1.000000
25%	0.000000	3.000000	3.000000
50%	1.000000	5.000000	4.000000
75%	1.000000	8.000000	5.000000
max	1.000000	10.000000	10.000000

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Continuous columns: ['Age', 'Anxiety Level (1-10)', 'Sleep Hours', 'Physical
Activity (hrs/week)']
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Age: 0 outliers (0.00%)

Anxiety Level (1-10): 694 outliers (6.31%)

[illegible]

9.0, 10.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 10.0, 9.0, 10.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0, 9.0, 10.0, 10.0, 10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 10.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0, 9.0, 10.0, 10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0, 10.0, 10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 10.0, 10.0, 9.0, 9.0, 10.0, 10.0, 10.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 9.0, 10.0, 10.0, 10.0, 9.0, 10.0, 10.0, 10.0, 9.0, 9.0, 10.0, 10.0, 10.0, 9.0, 9.0, 10.0, 10.0, 9.0, 10.0, 9.0]

Sleep Hours: 28 outliers (0.25%)

Bounds: [2.15, 11.35]

Outlier values: [11.8, 11.4, 2.1, 11.6, 12.2, 11.7, 1.9, 11.4, 2.0, 12.5, 1.8, 11.4, 11.4, 1.7, 11.5, 1.8, 1.6, 11.9, 11.9, 1.3, 12.1, 11.7, 0.2, 11.4, 11.4, 12.5, 11.4, 11.7]

Physical Activity (hrs/week): 30 outliers (0.27%)

Bounds: [-2.80, 8.40]

Outlier values: [10.8, 9.2, 8.5, 9.8, 10.2, 8.6, 8.5, 8.5, 10.0, 9.3, 8.6, 8.7, 9.6, 8.5, 9.5, 11.2, 9.8, 8.5, 9.0, 8.5, 9.4, 8.7, 9.9, 8.6, 9.1, 8.6, 9.0, 8.9, 9.0, 8.9]

ZSCORE METHOD OUTLIER RESULTS

Age: 0 outliers (0.00%)

Max ZScore: 1.77

Anxiety Level (1-10): 0 outliers (0.00%)

Max ZScore: 2.76

Sleep Hours: 17 outliers (0.15%)

Max ZScore: 4.07

Outlier values: [11.8, 12.2, 11.7, 1.9, 12.5, 1.8, 1.7, 1.8, 1.6, 11.9, 11.9, 1.3, 12.1, 11.7, 0.2, 12.5, 11.7]

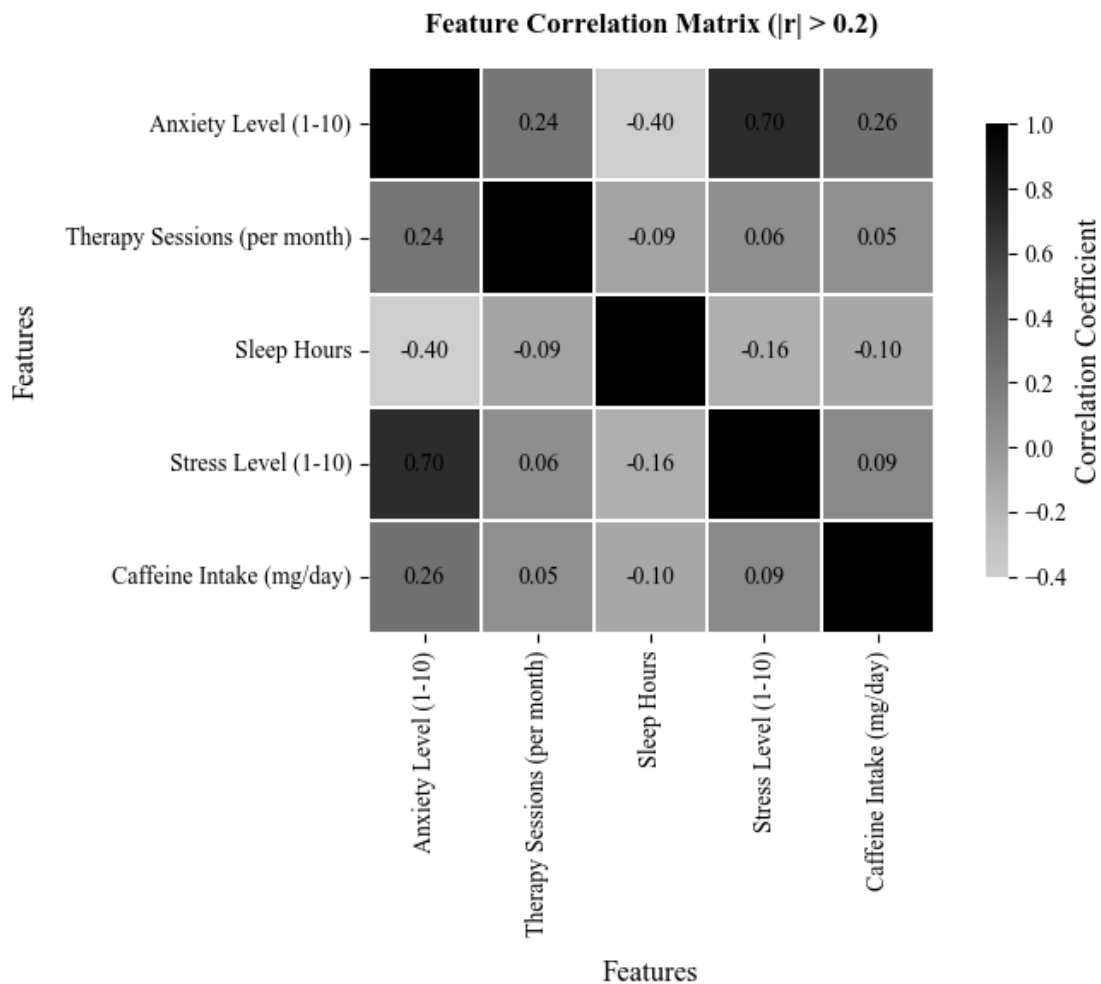
Physical Activity (hrs/week): 30 outliers (0.27%)

Max ZScore: 4.51

Outlier values: [10.8, 9.2, 8.5, 9.8, 10.2, 8.6, 8.5, 8.5, 10.0, 9.3, 8.6, 8.7, 9.6, 8.5, 9.5, 11.2, 9.8, 8.5, 9.0, 8.5, 9.4, 8.7, 9.9, 8.6, 9.1, 8.6, 9.0, 8.9, 9.0, 8.9]

HIGH CORRELATIONS ABS > 0.2

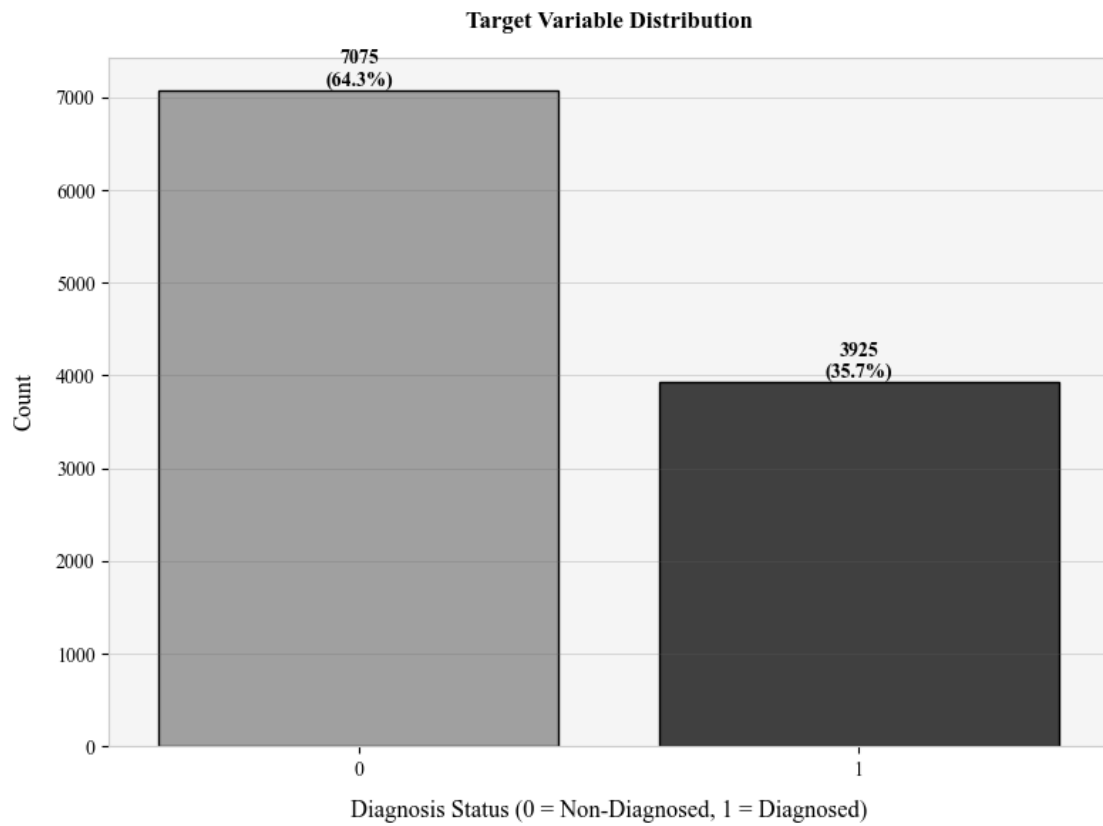
	Feature 1	Feature 2	Correlation
	Stress Level (1-10)	Anxiety Level (1-10)	0.699490
	Caffeine Intake (mg/day)	Anxiety Level (1-10)	0.256051
	Therapy Sessions (per month)	Anxiety Level (1-10)	0.240451
	Sleep Hours	Anxiety Level (1-10)	-0.401197

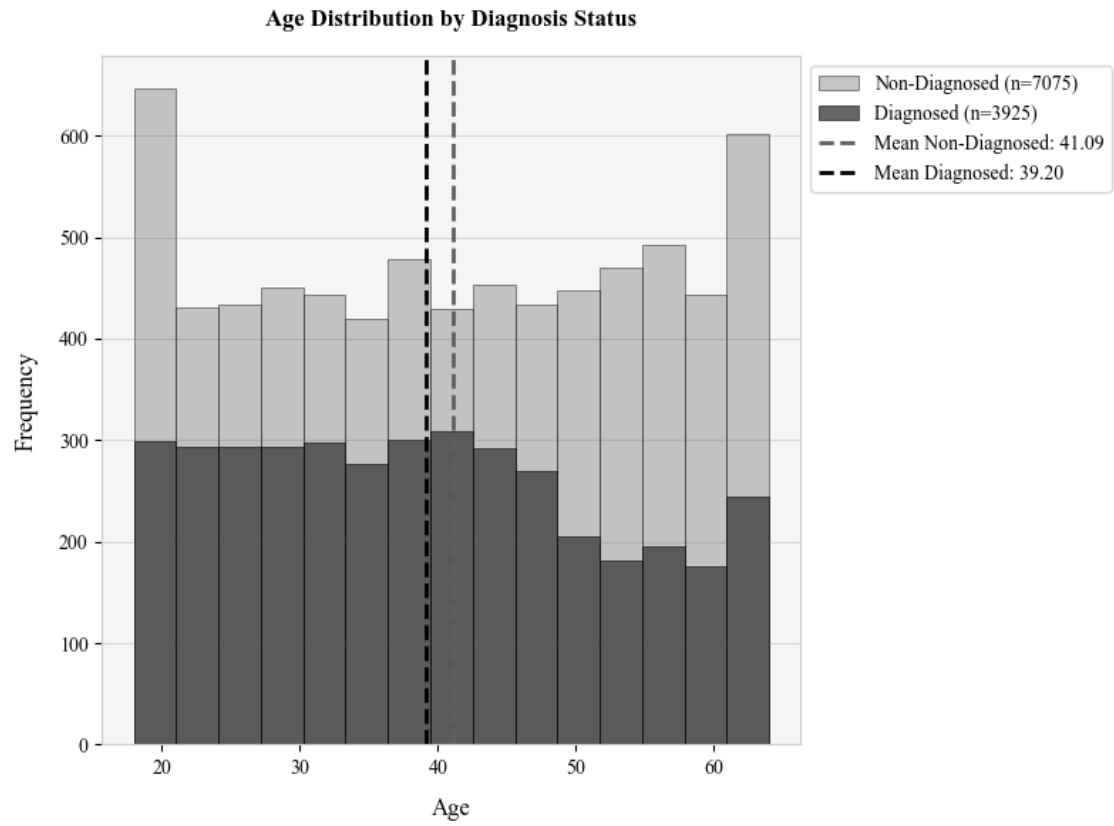


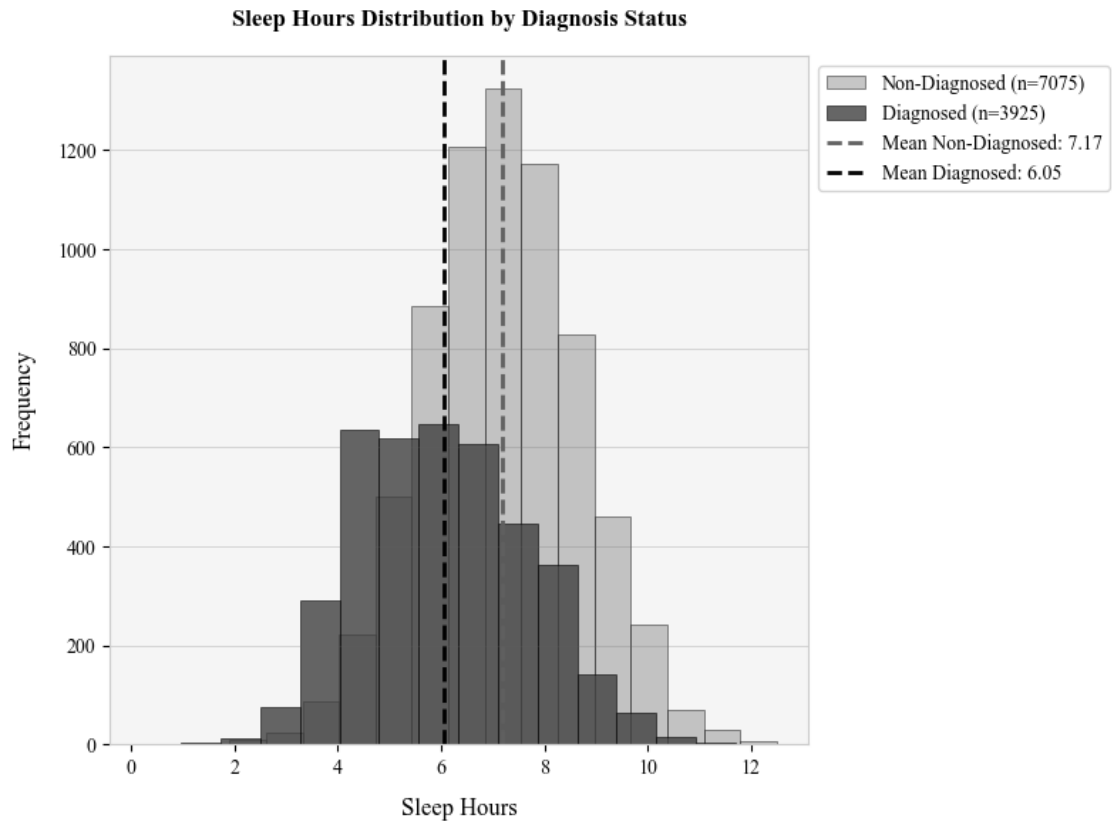
/tmp/ipykernel_3126/2562788784.py:209: FutureWarning:

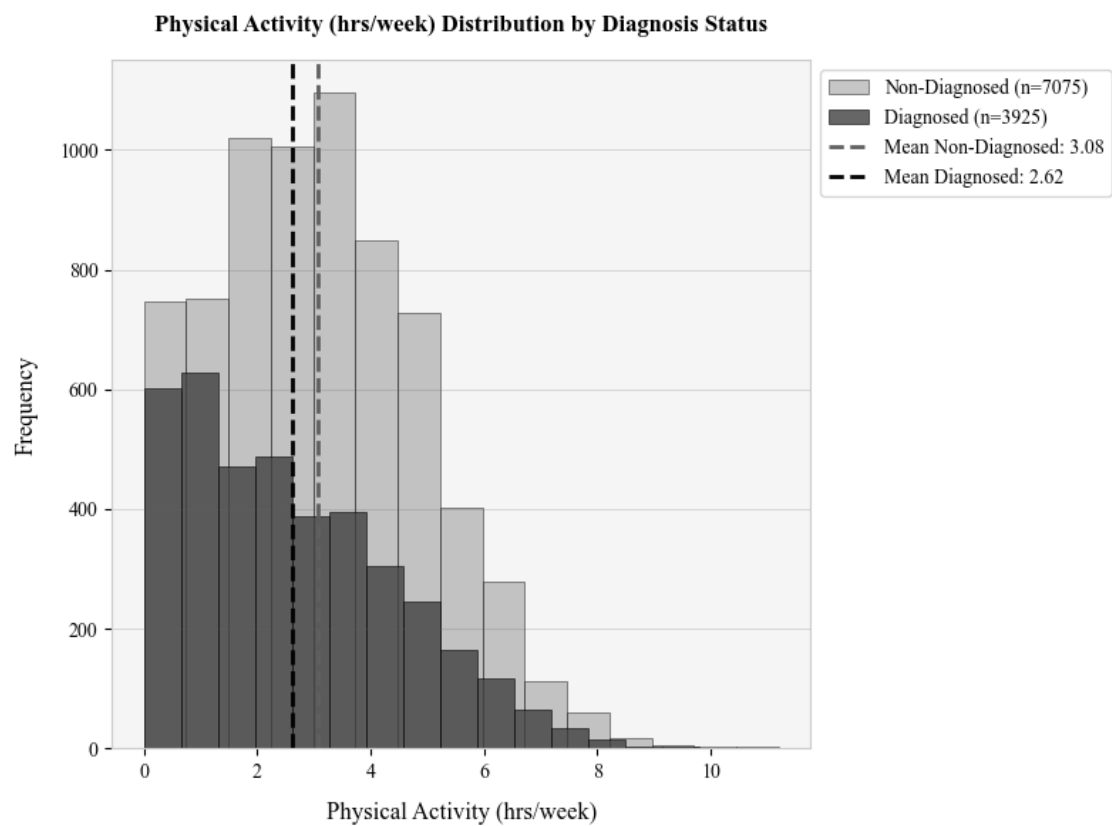
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

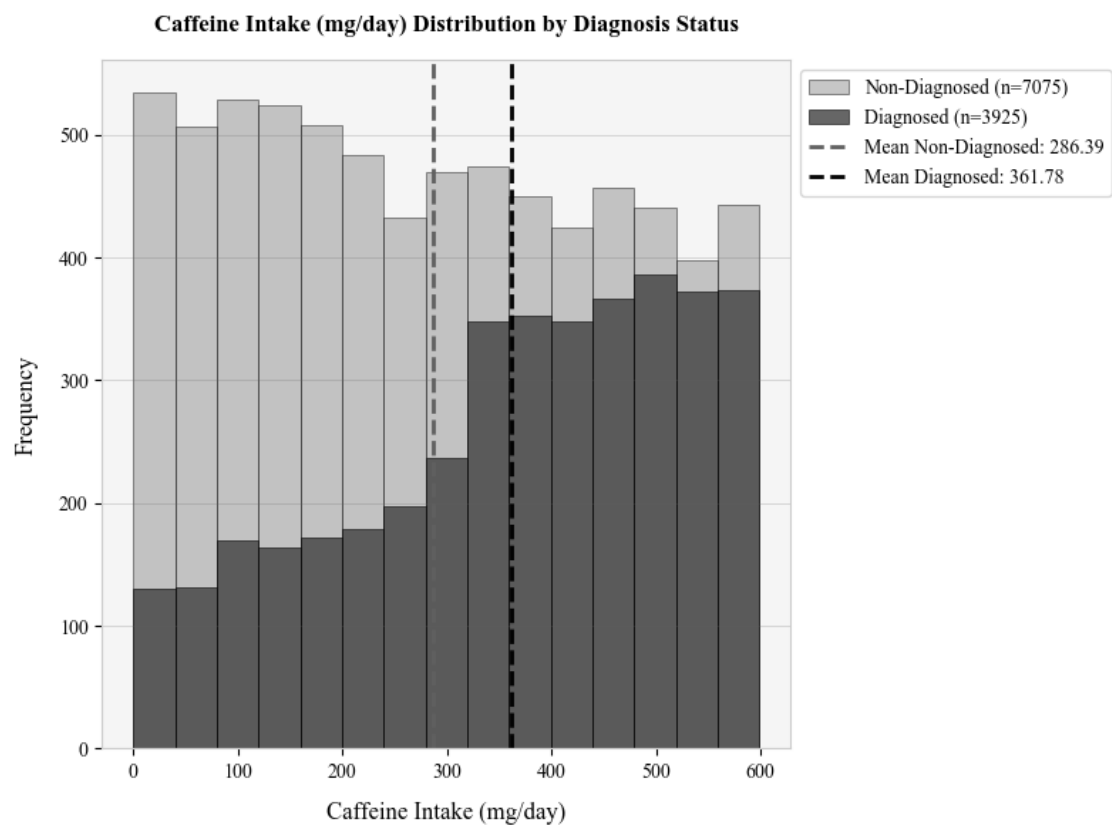
```
bars = sns.countplot(x=data['Diagnosed'], palette=['#a0a0a0', '#404040'],
```

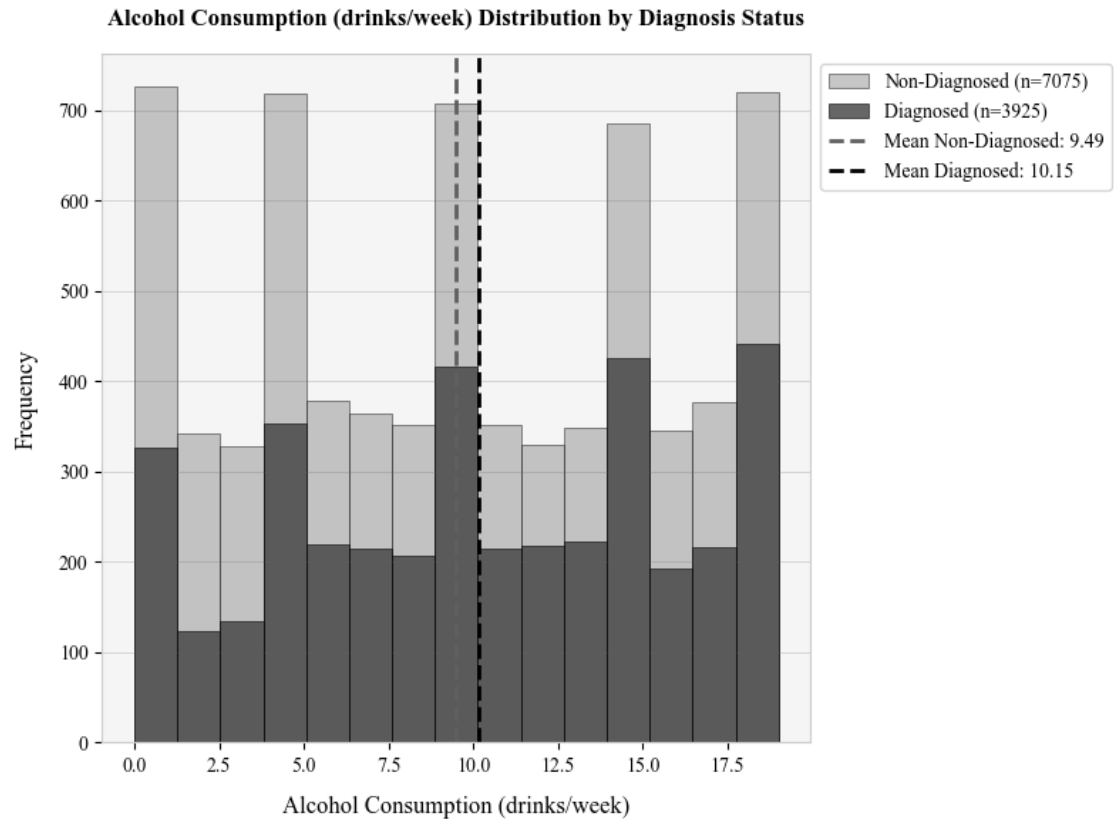


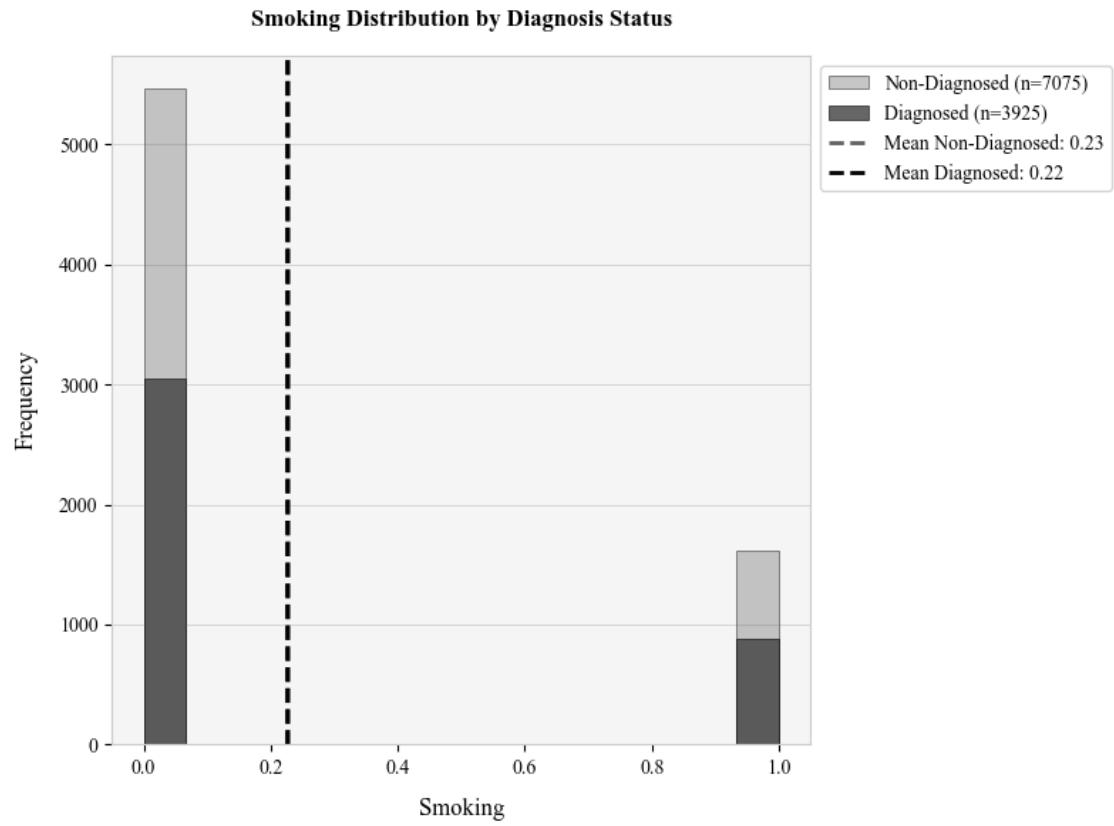


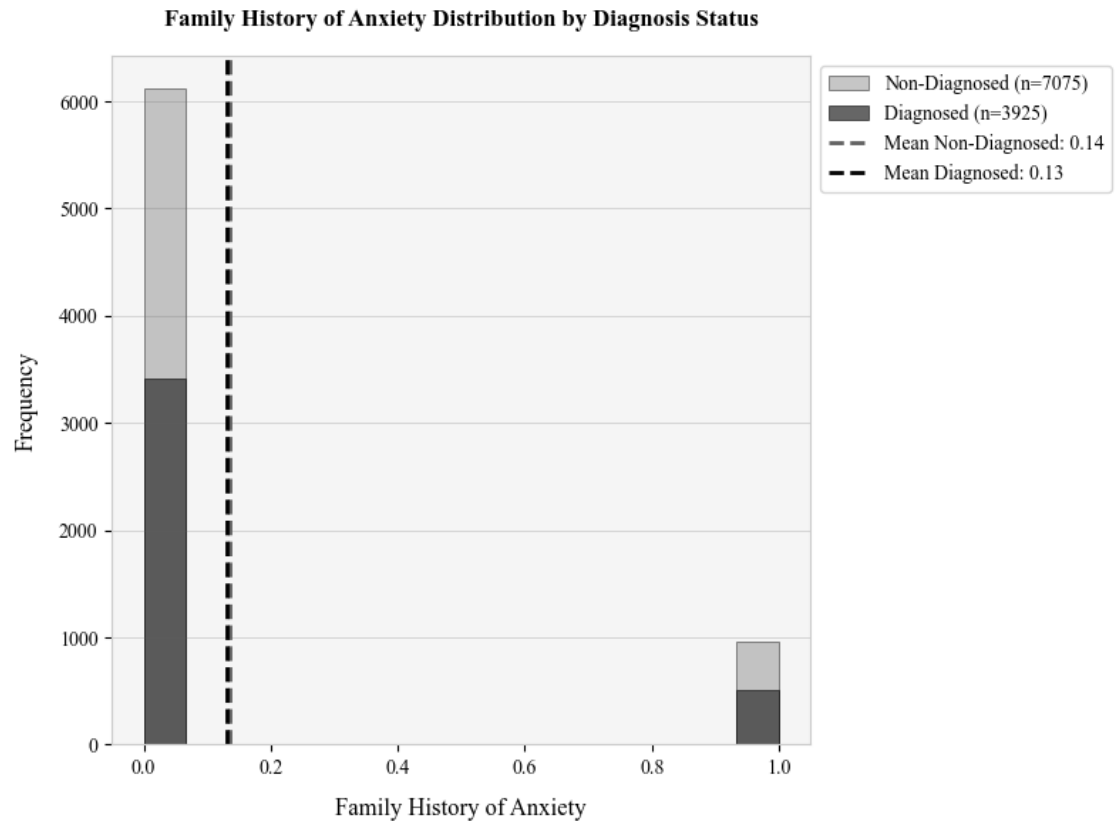


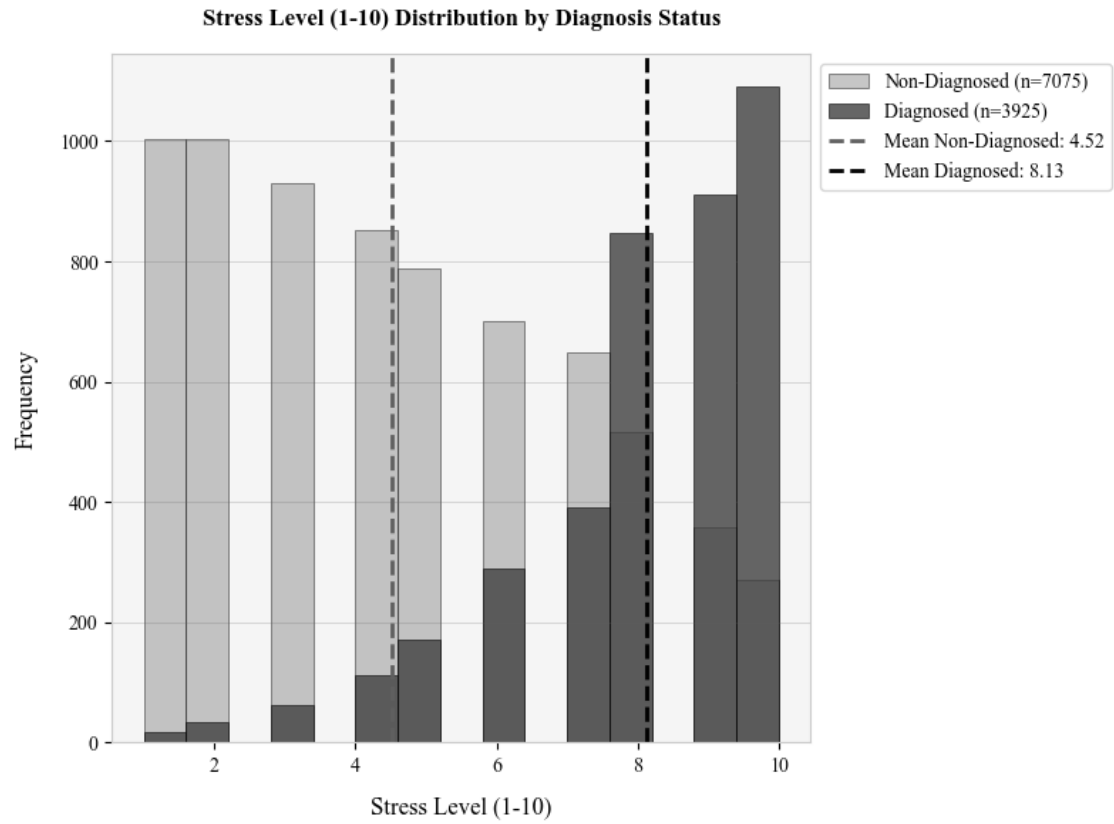


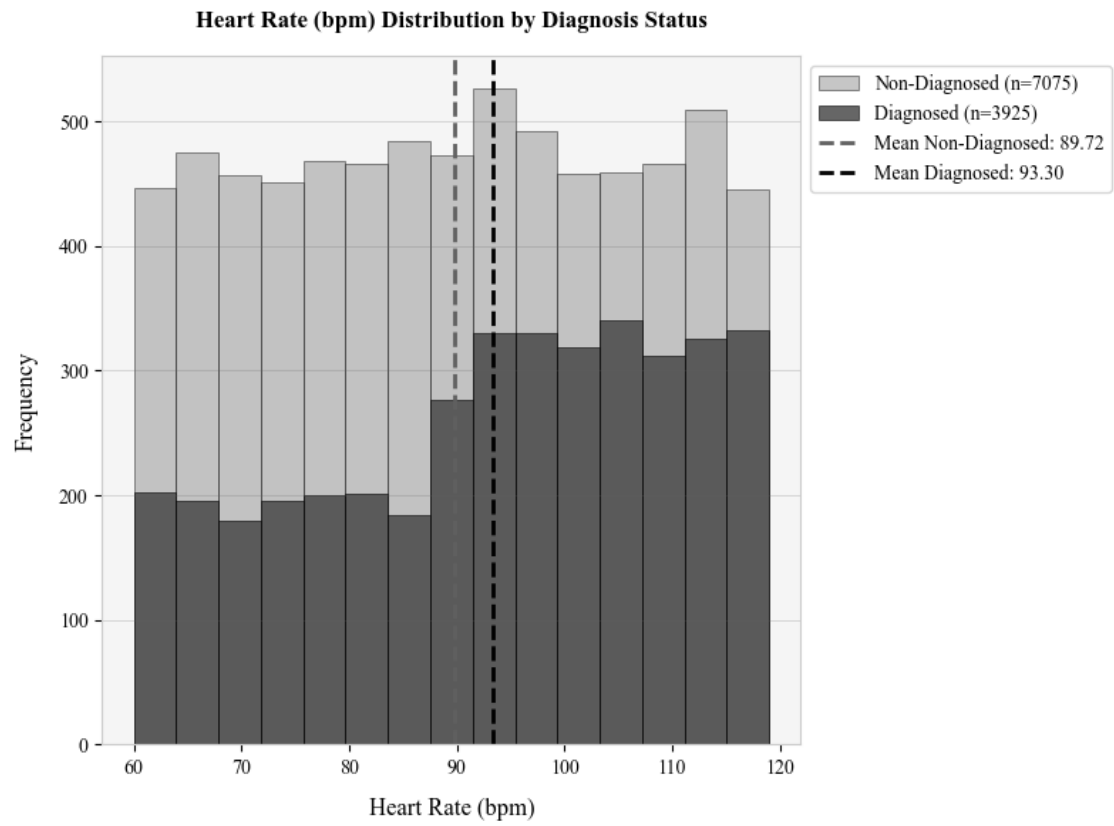


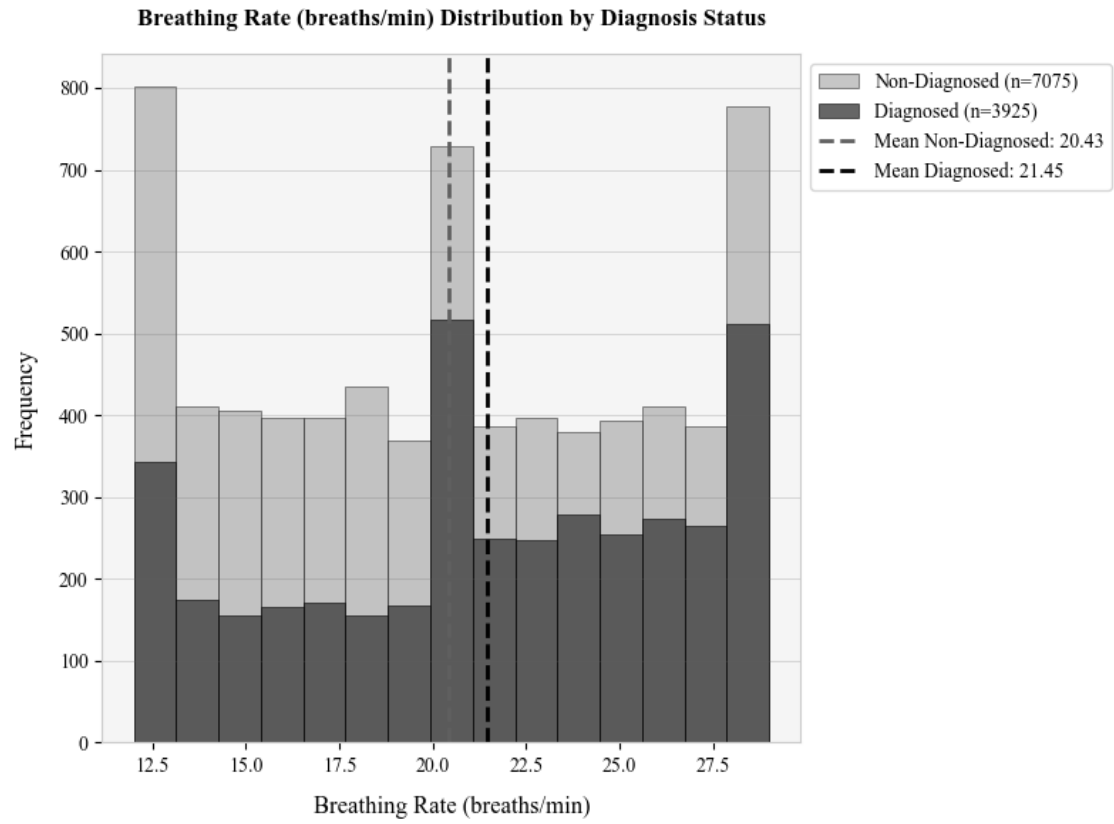


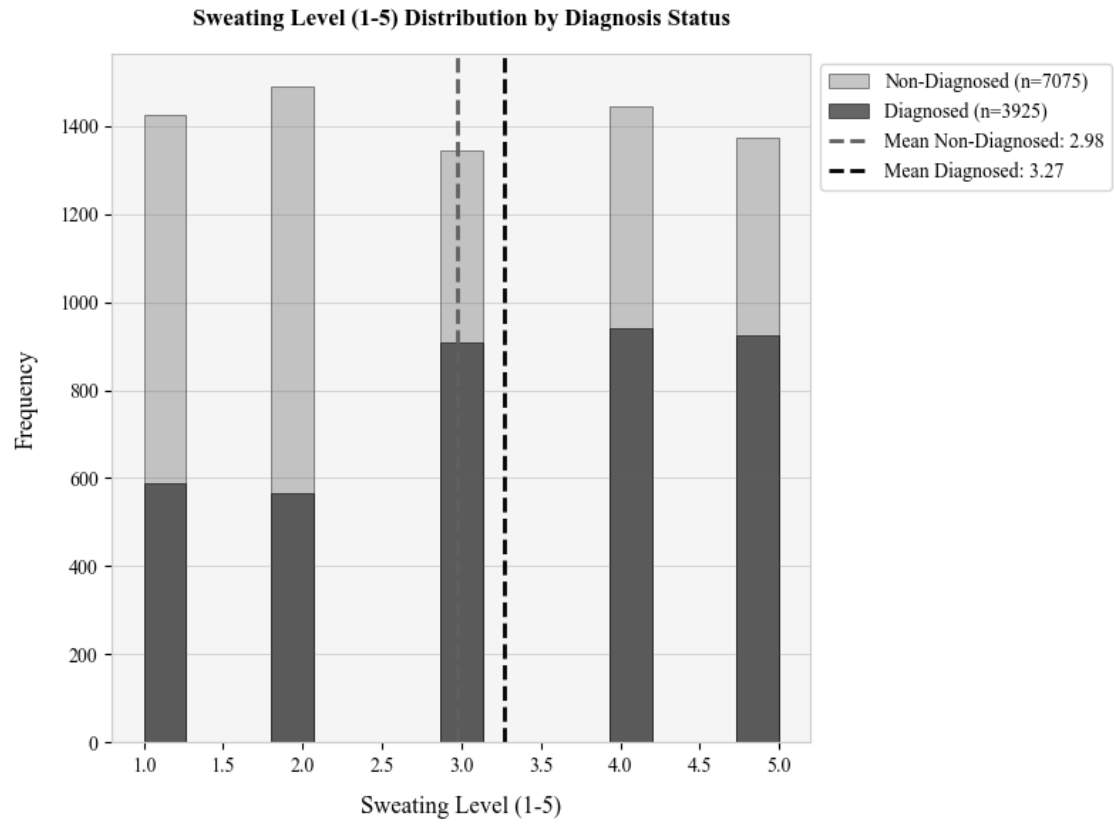


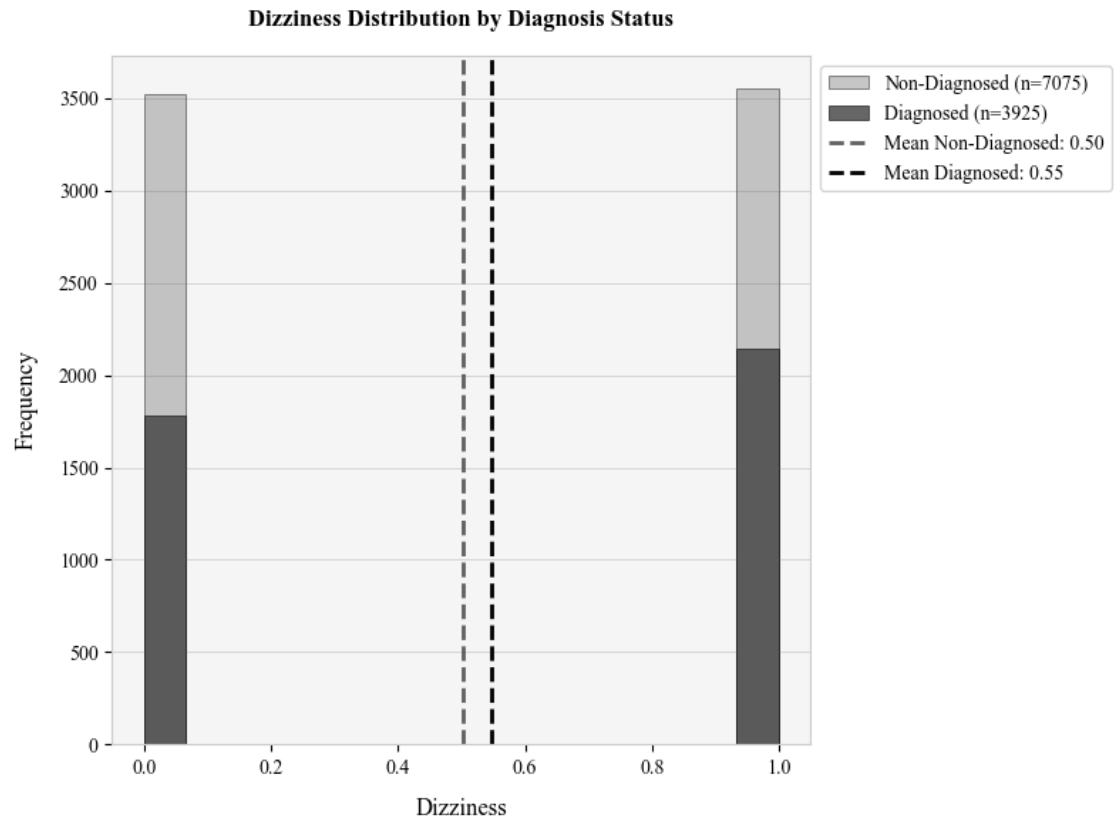


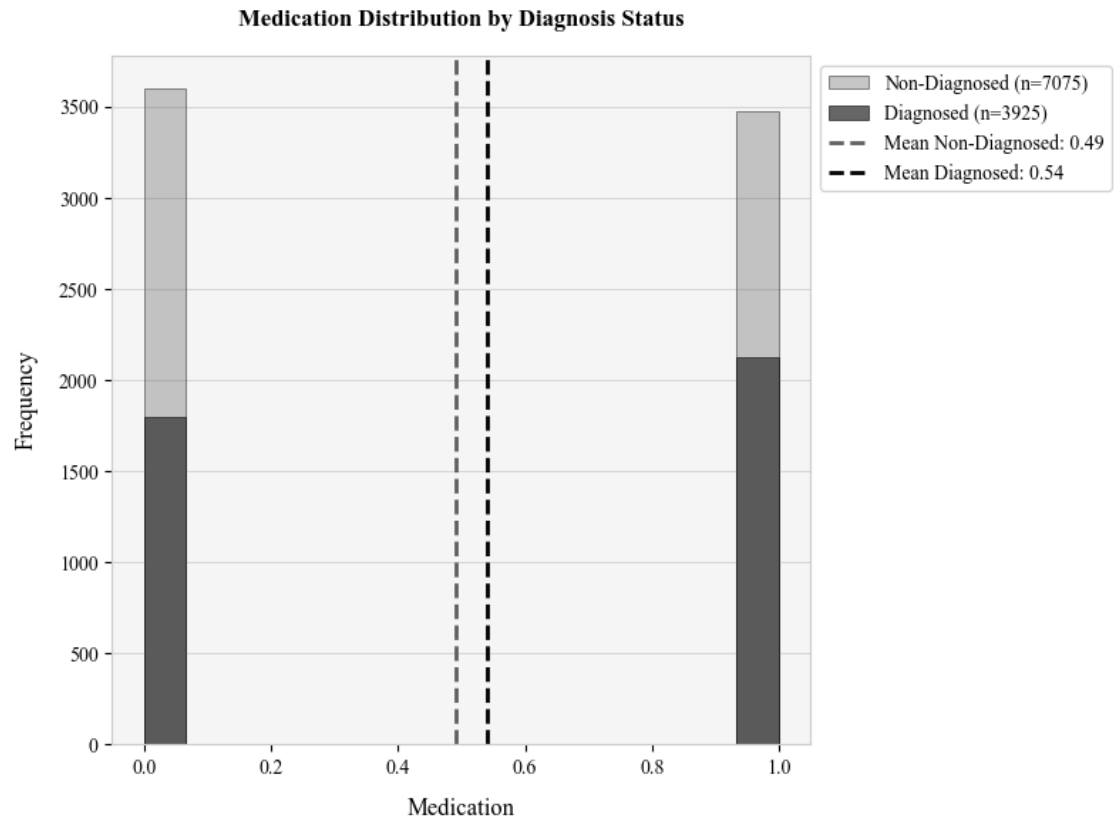


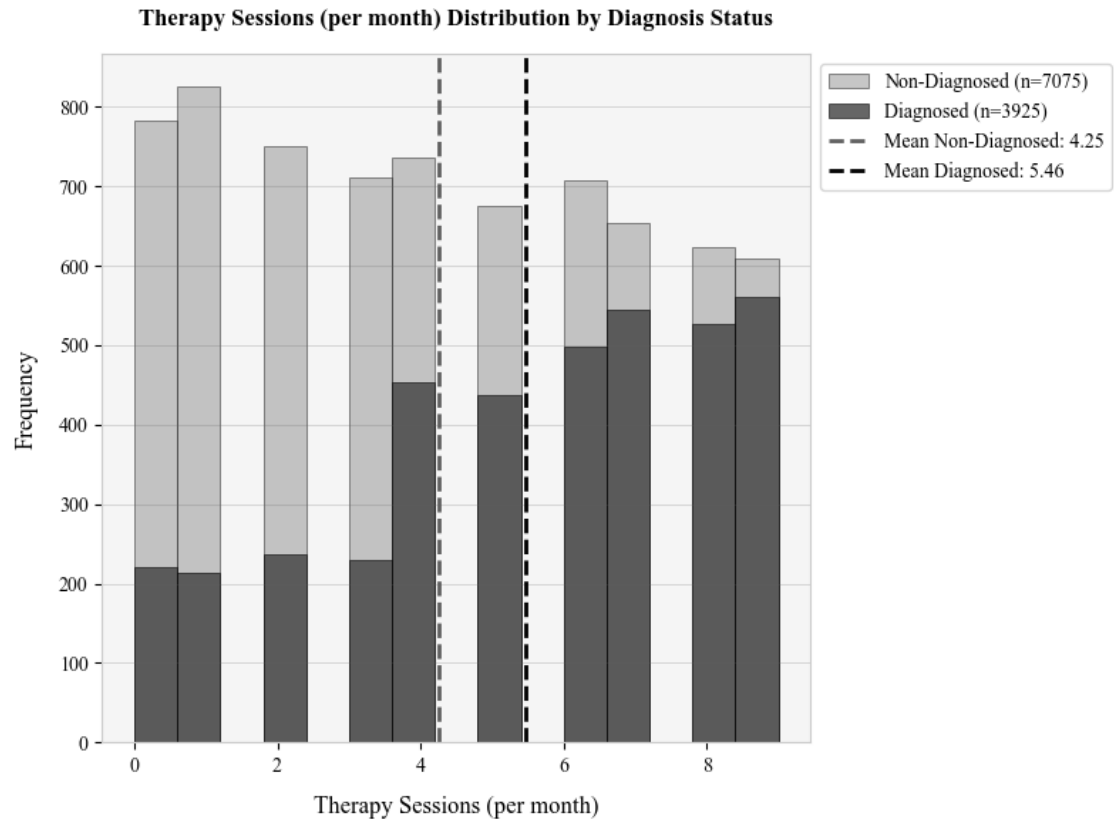


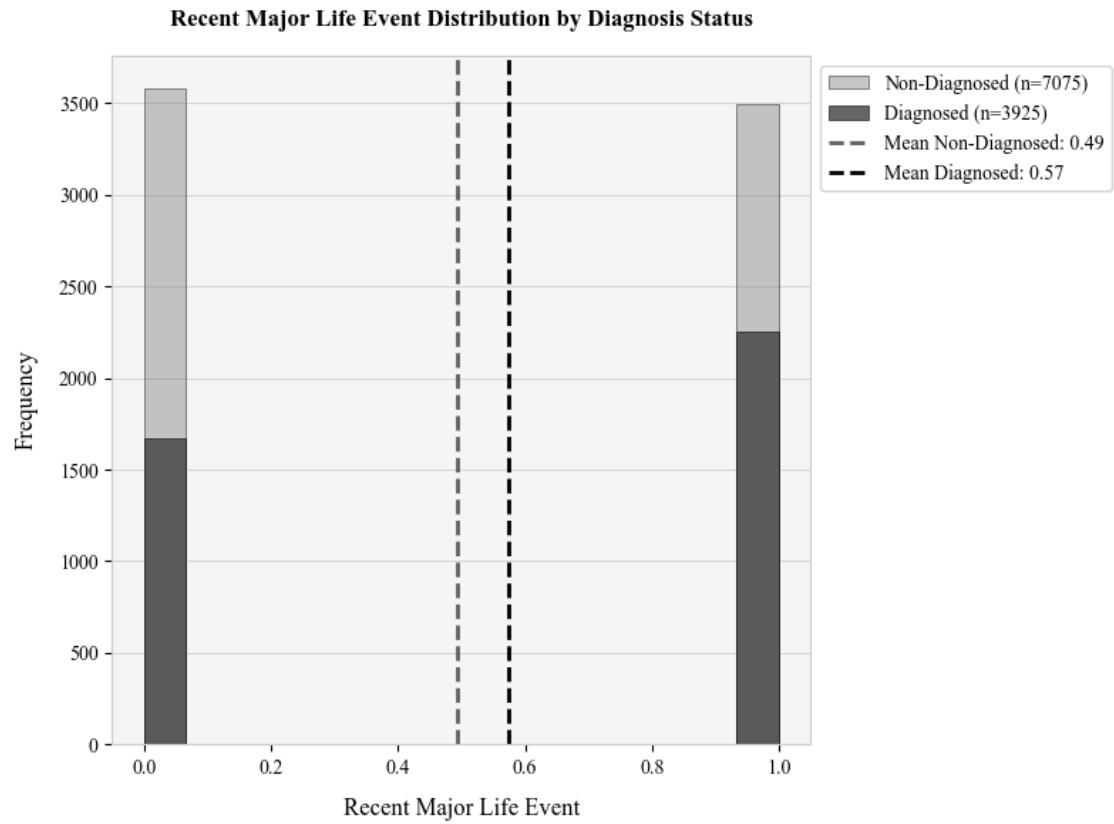


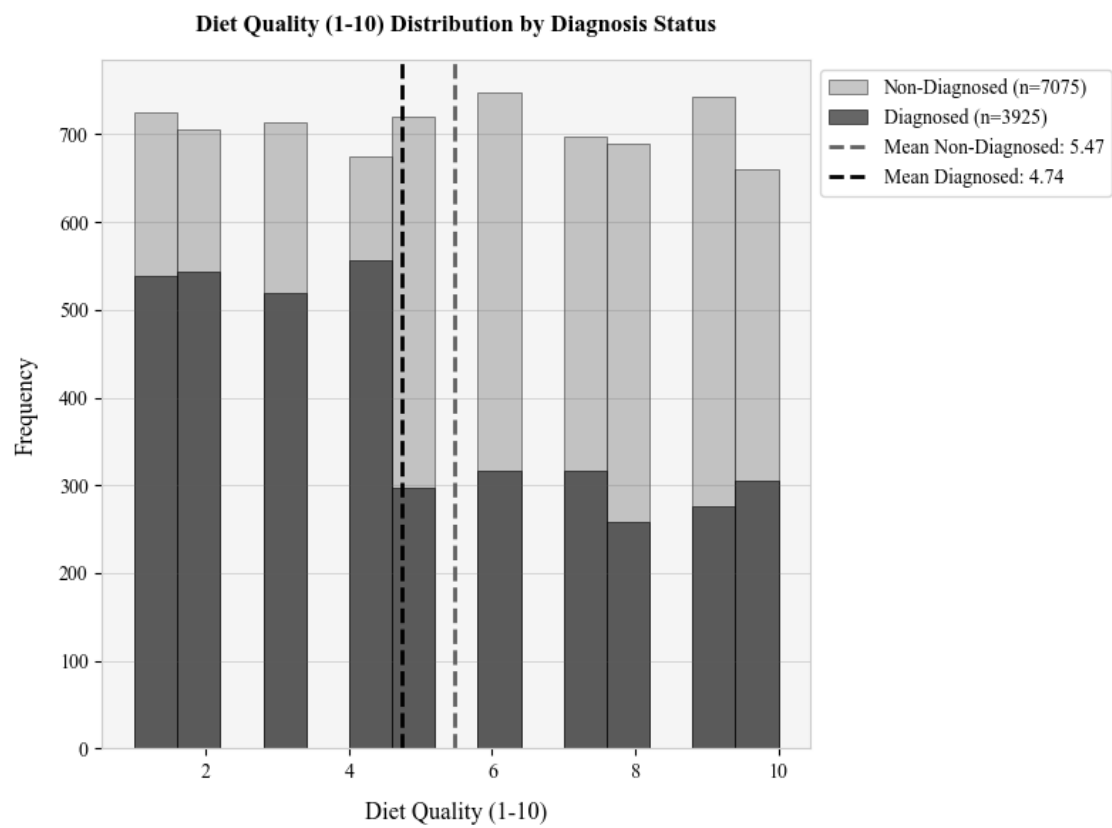


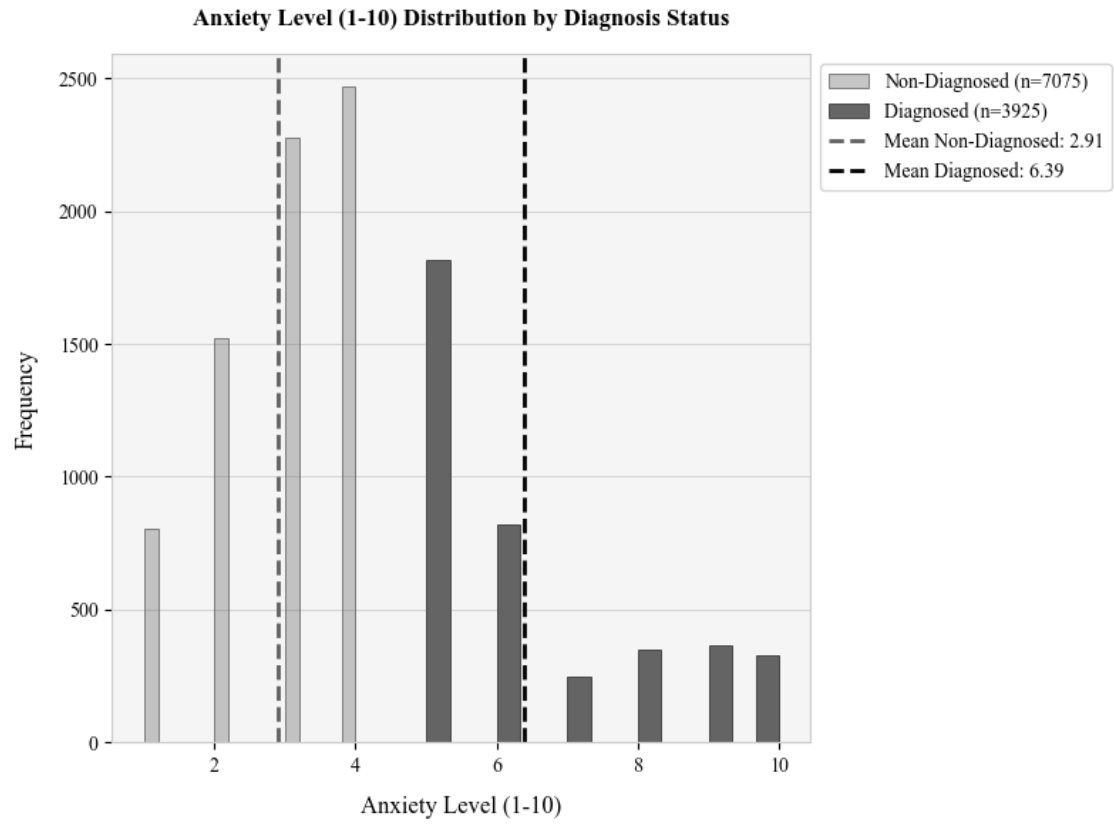


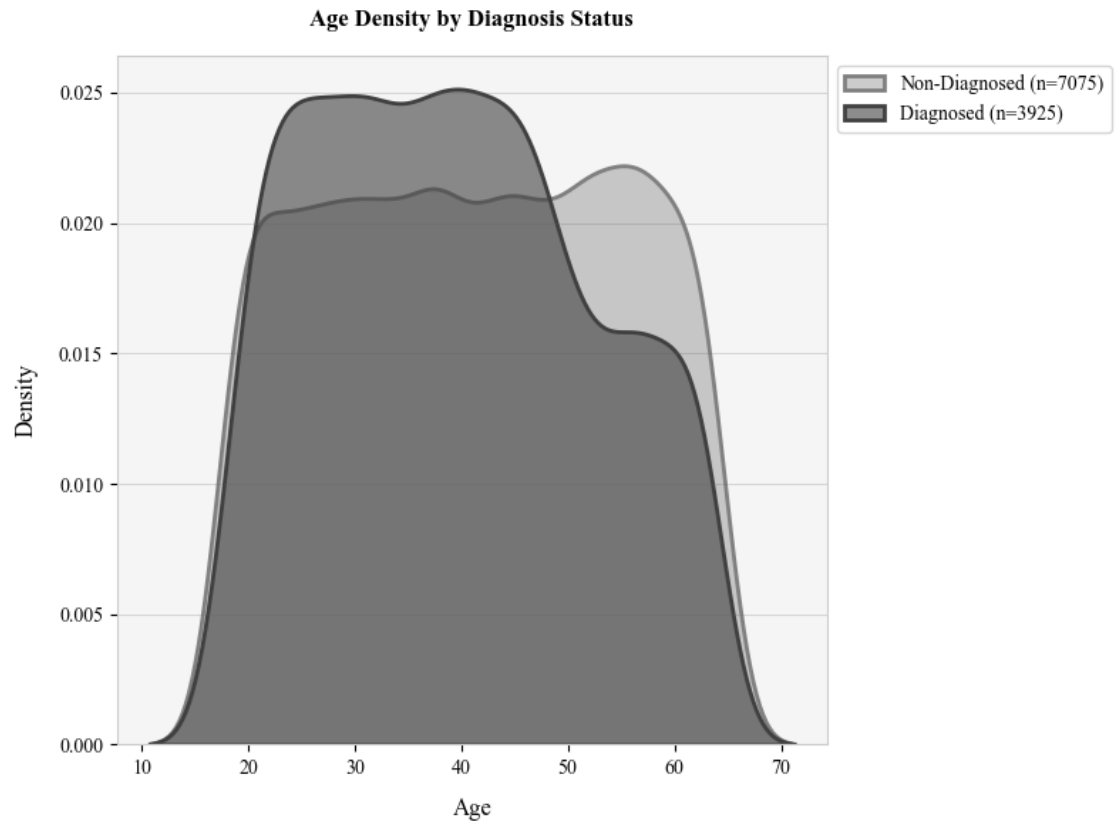


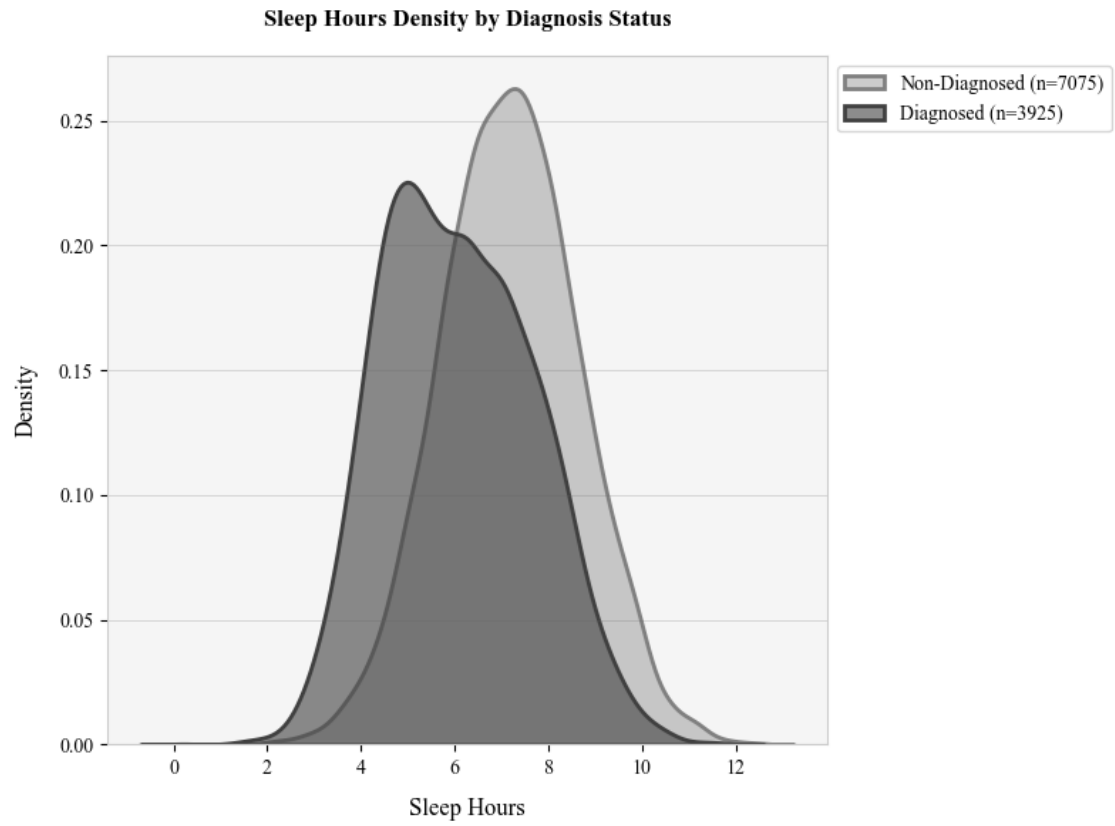


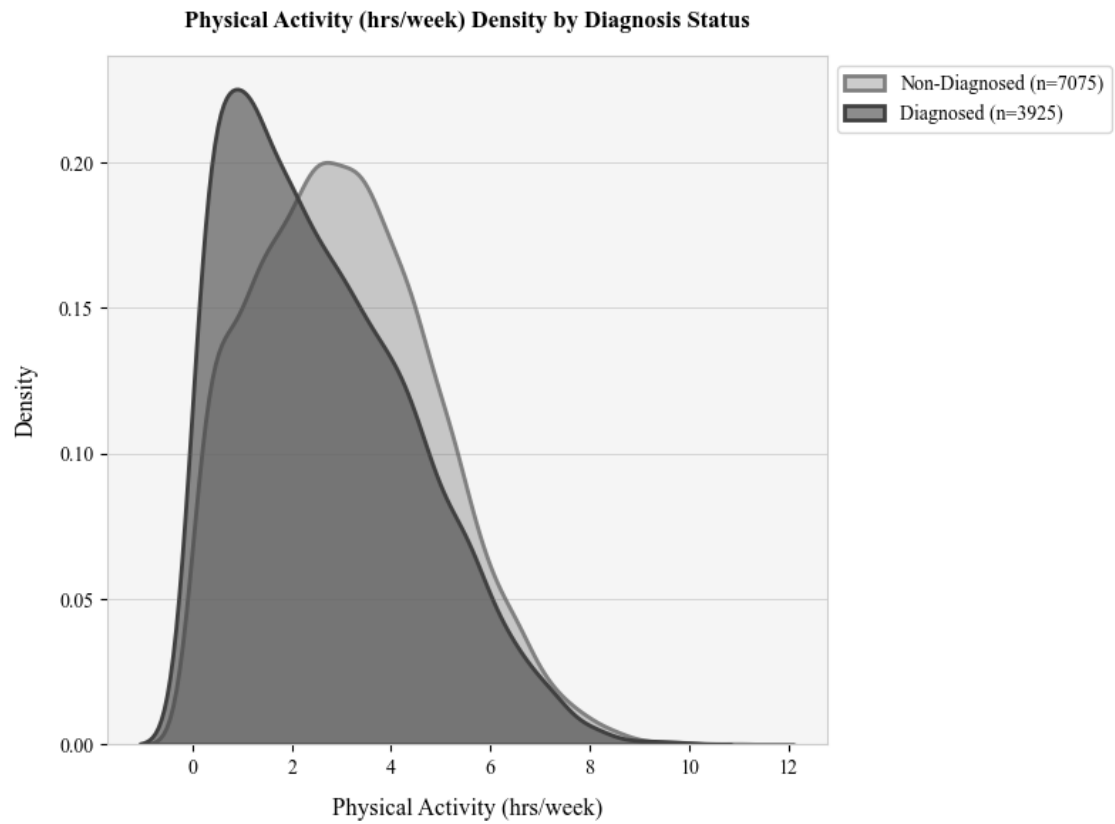


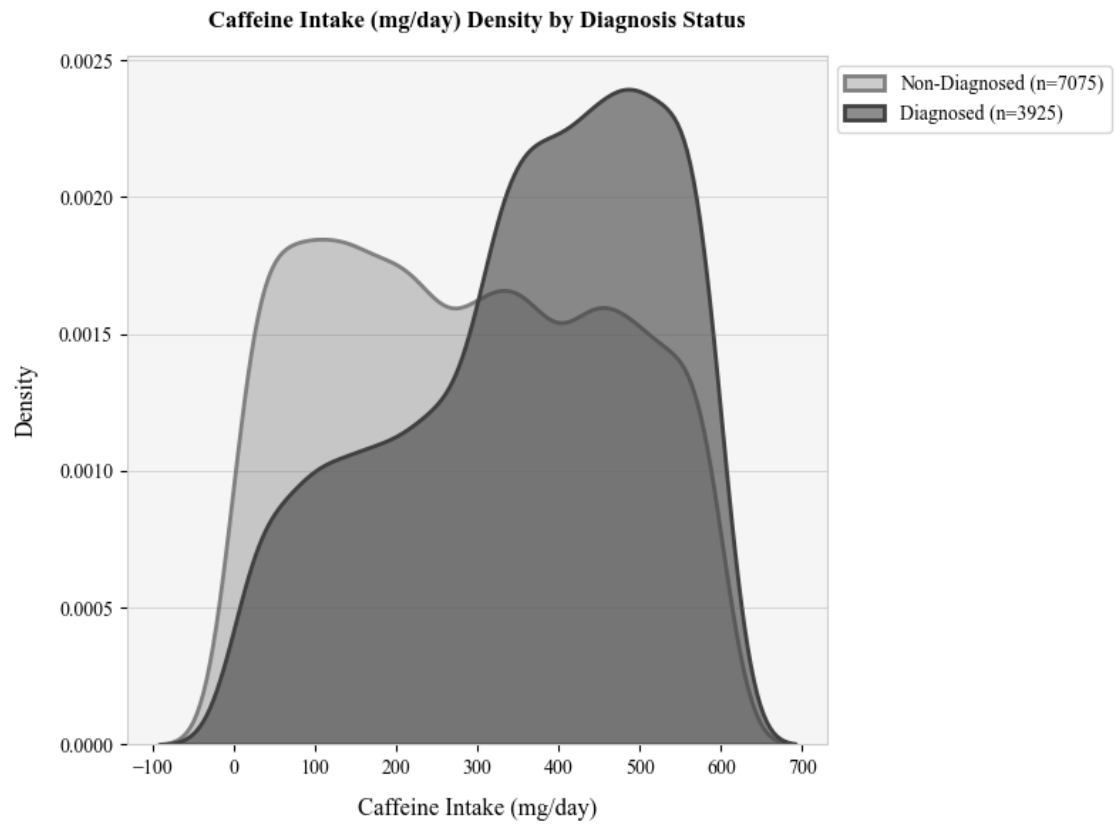


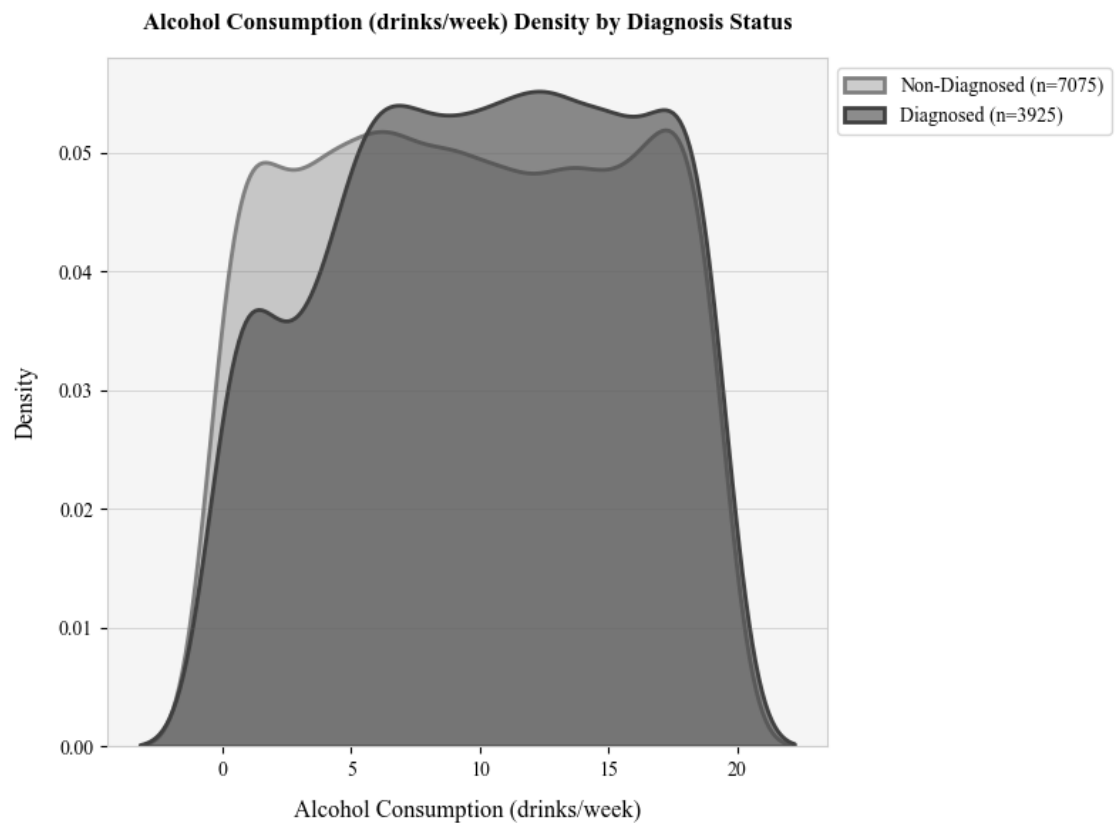


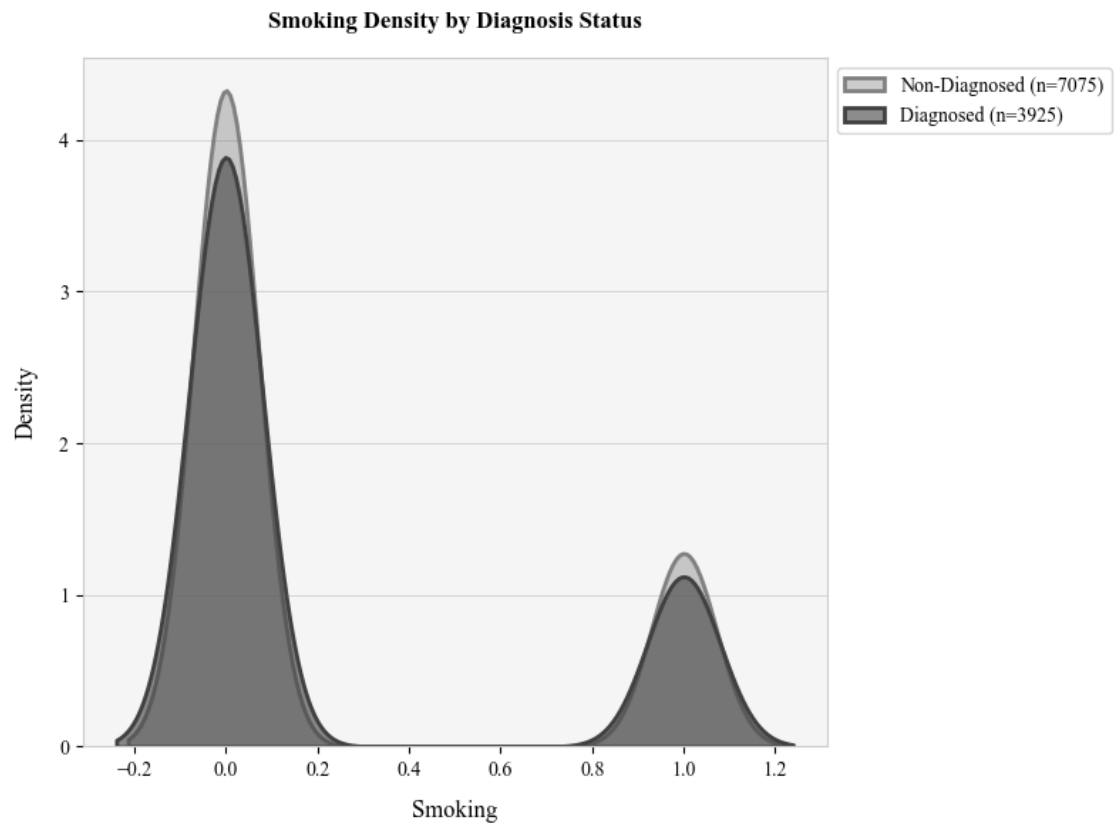


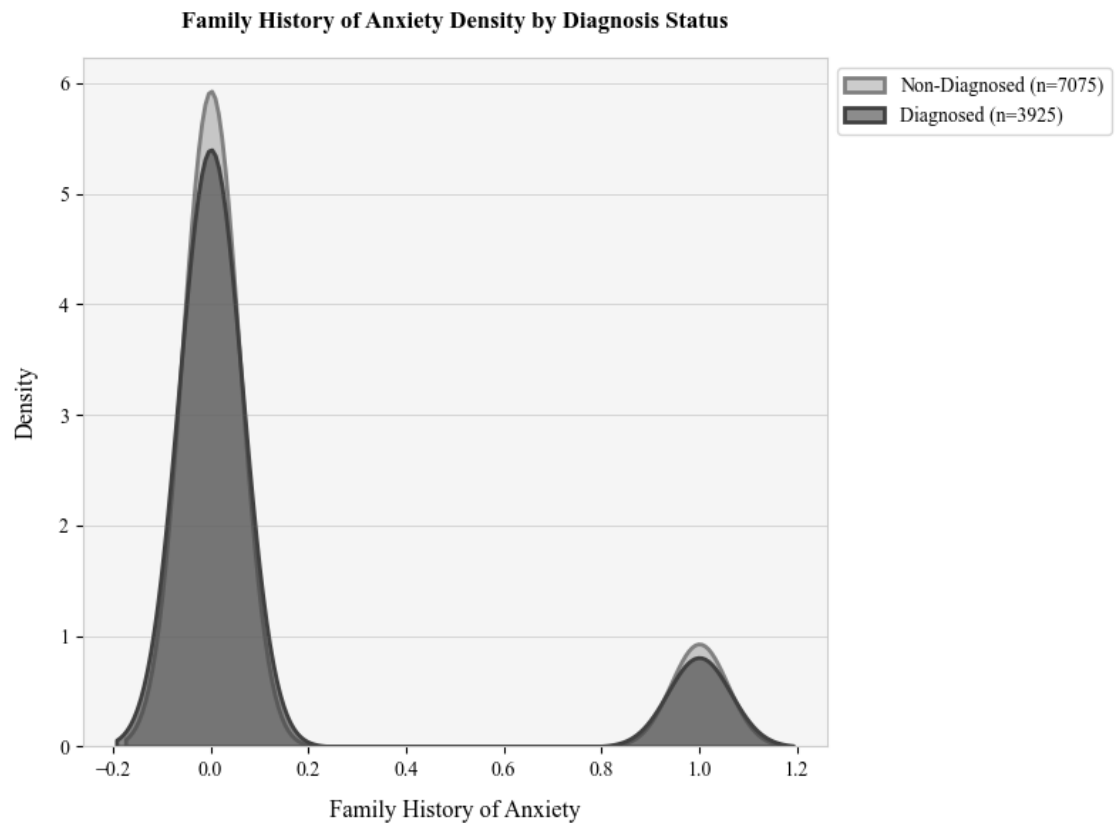


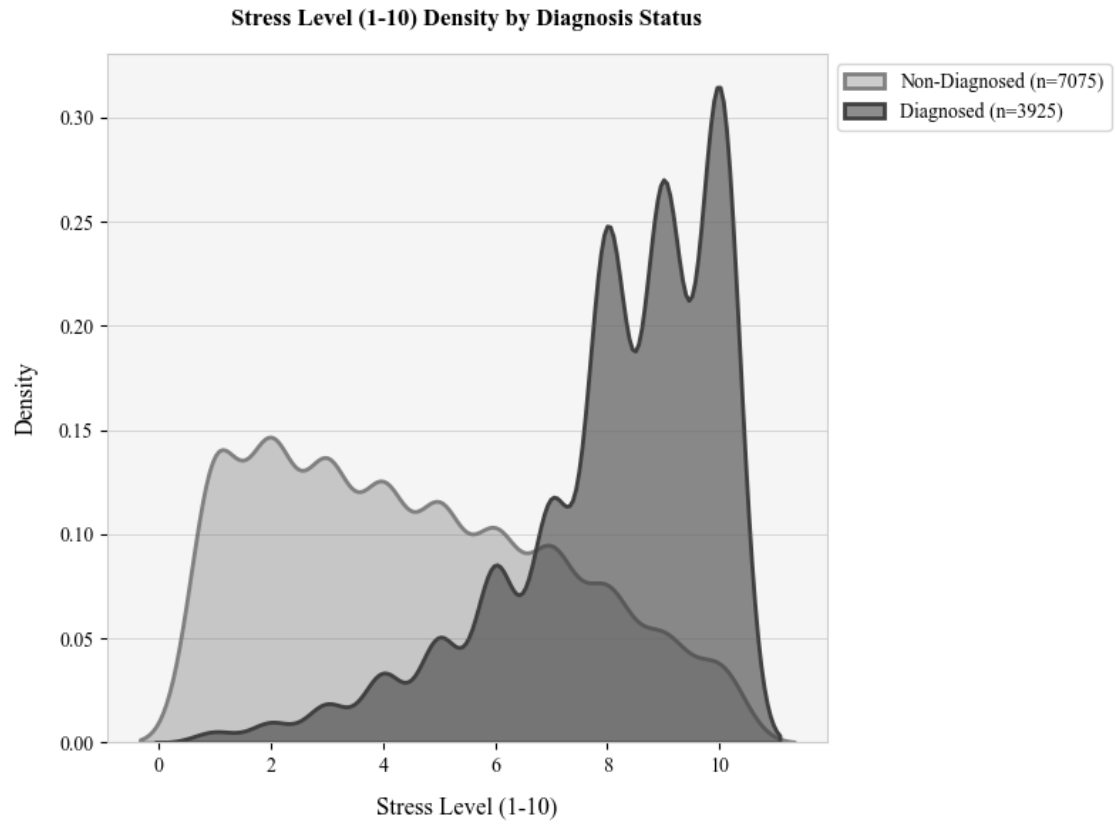


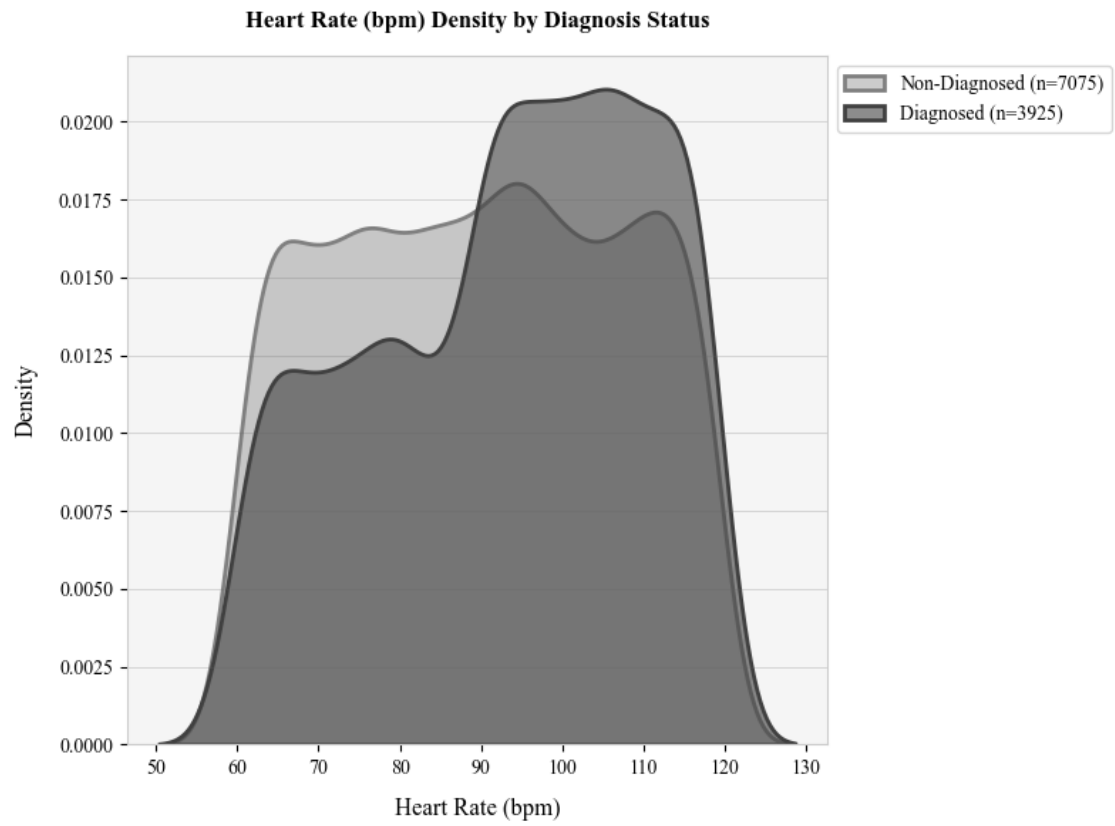


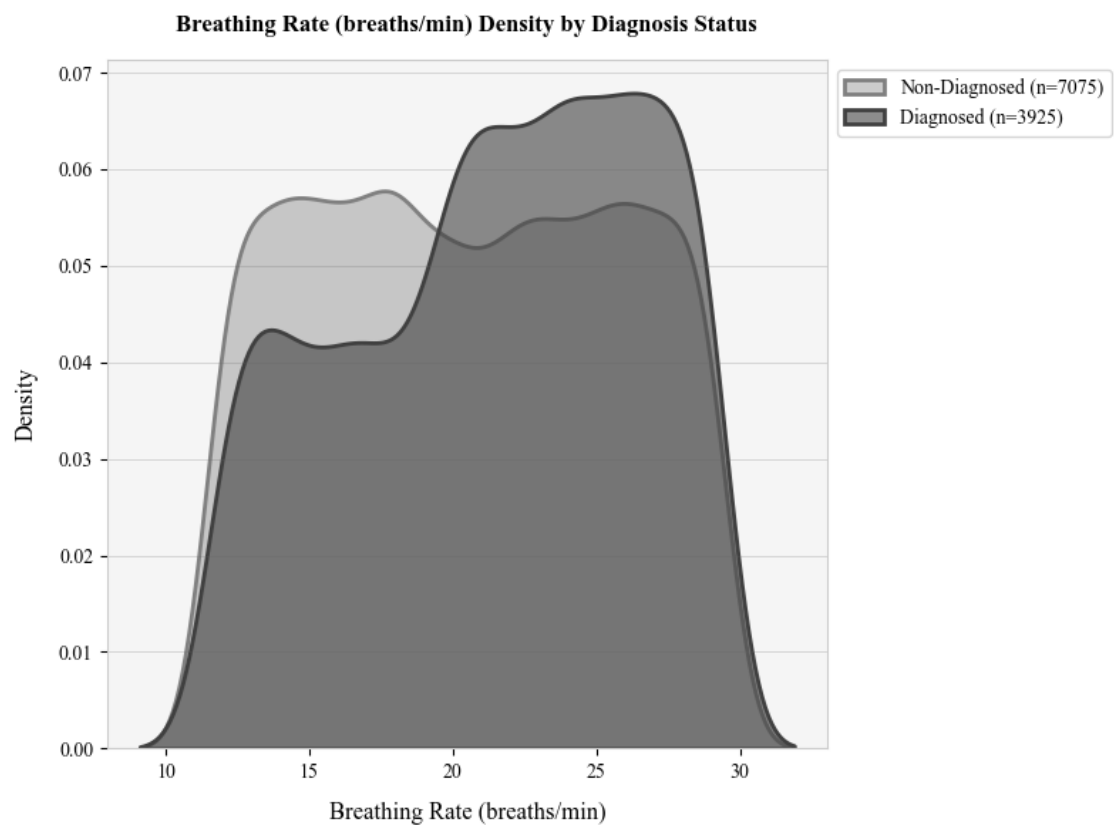


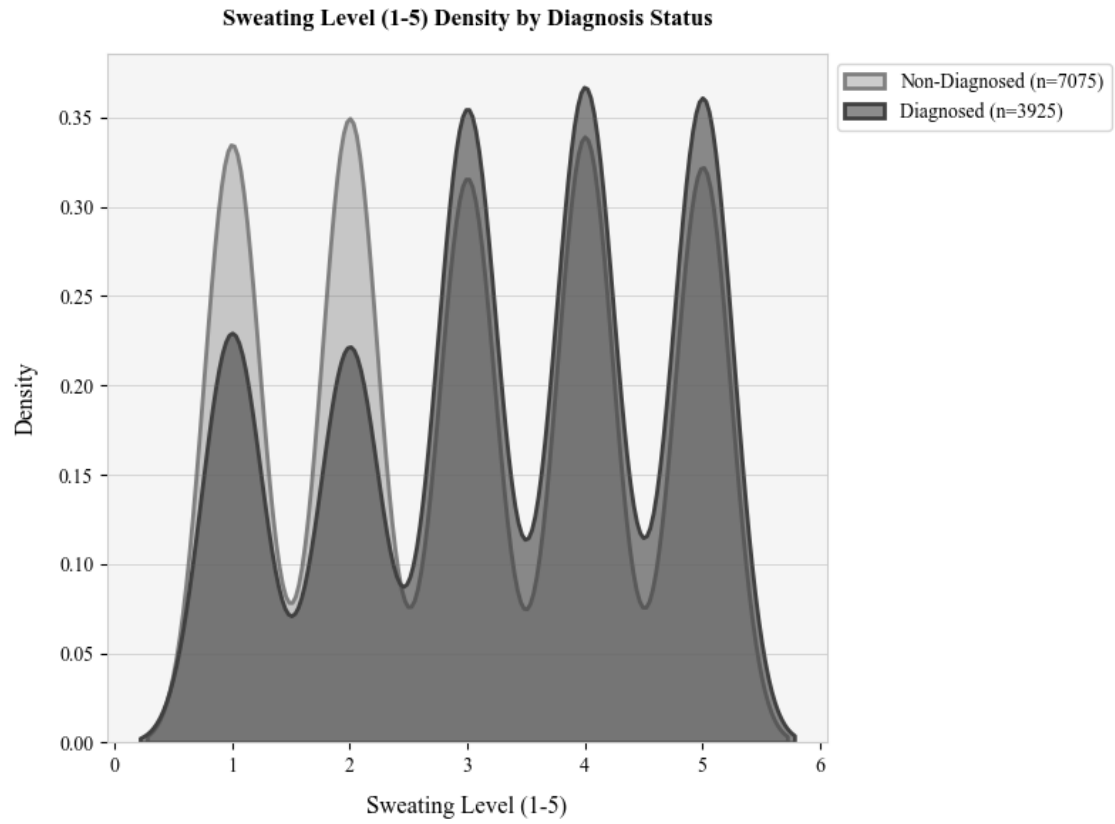


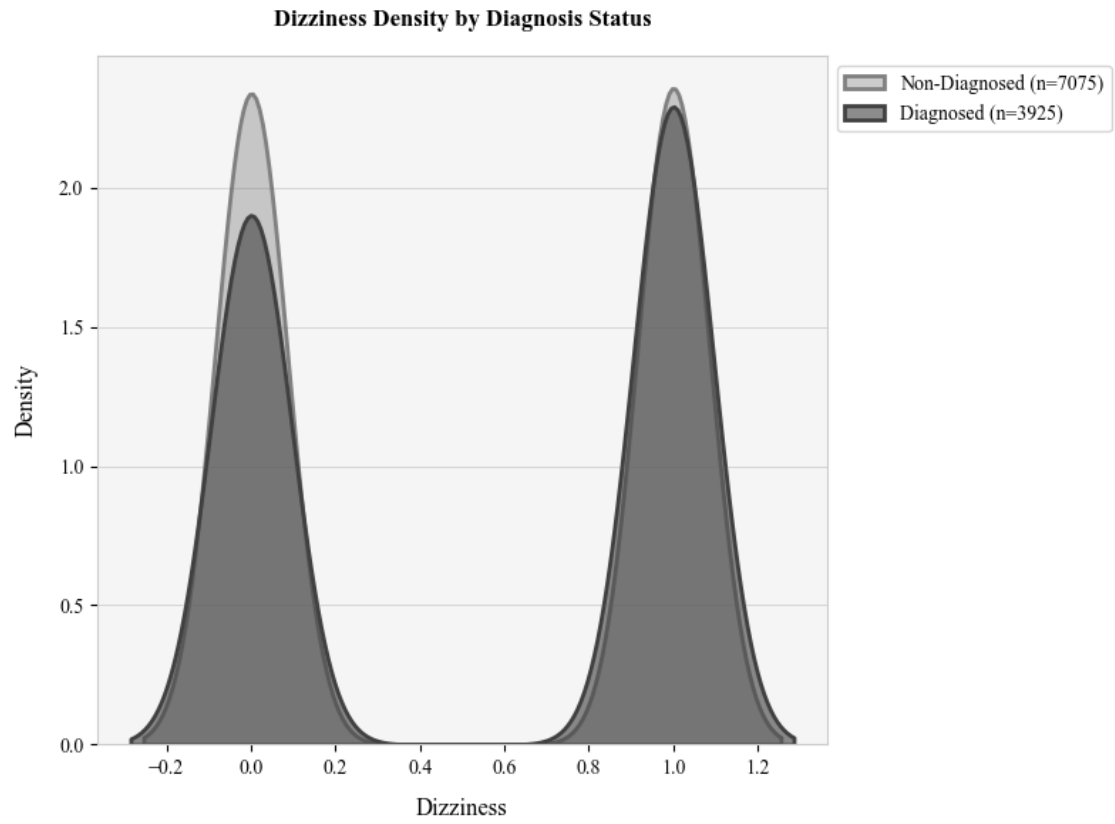


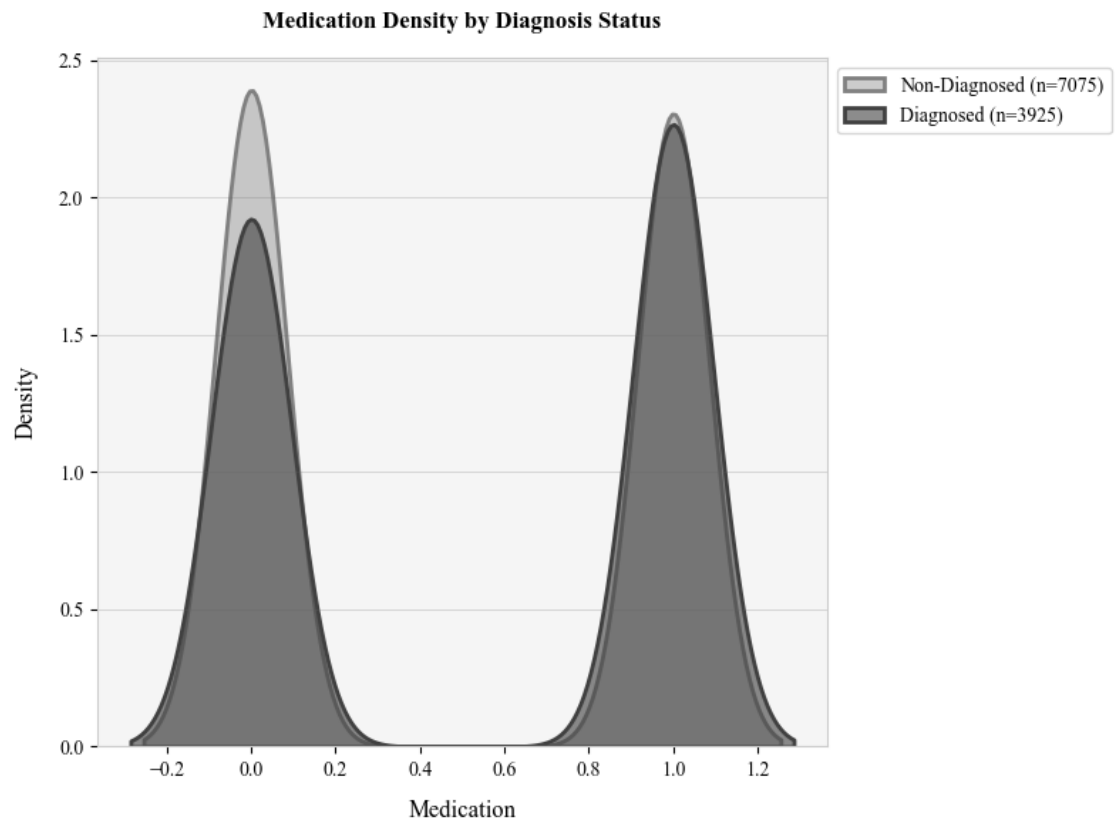


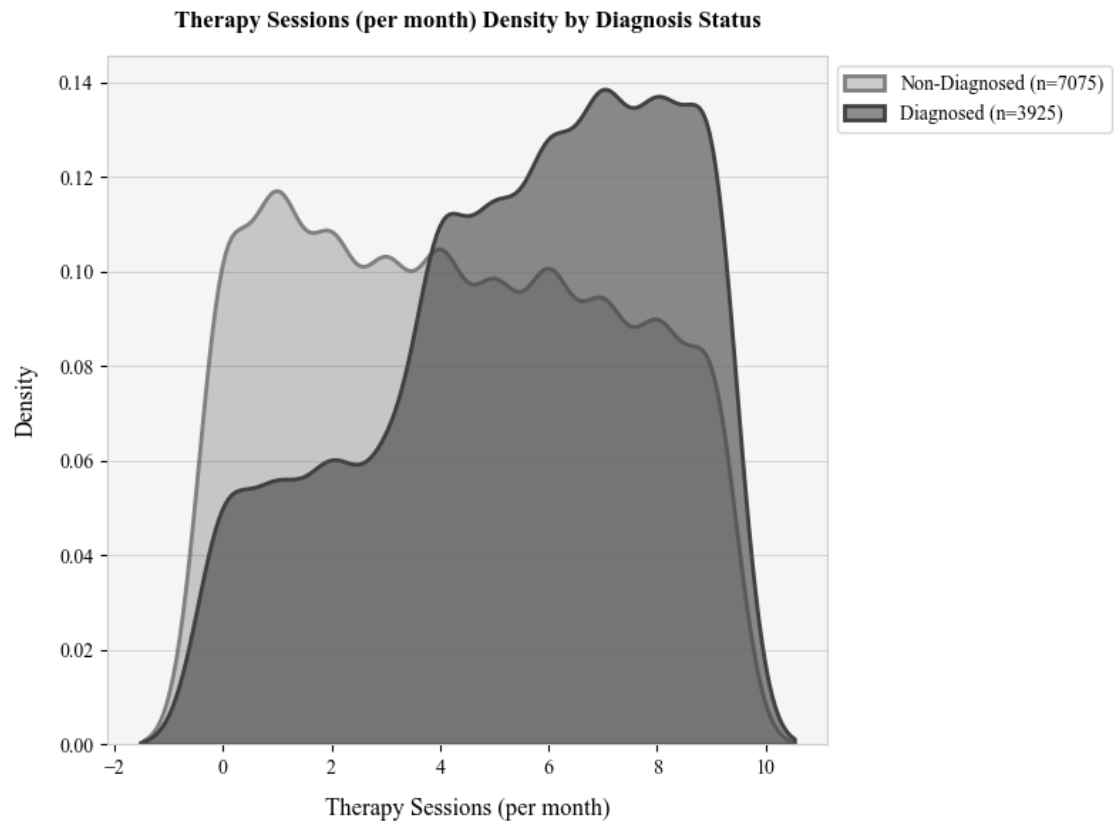


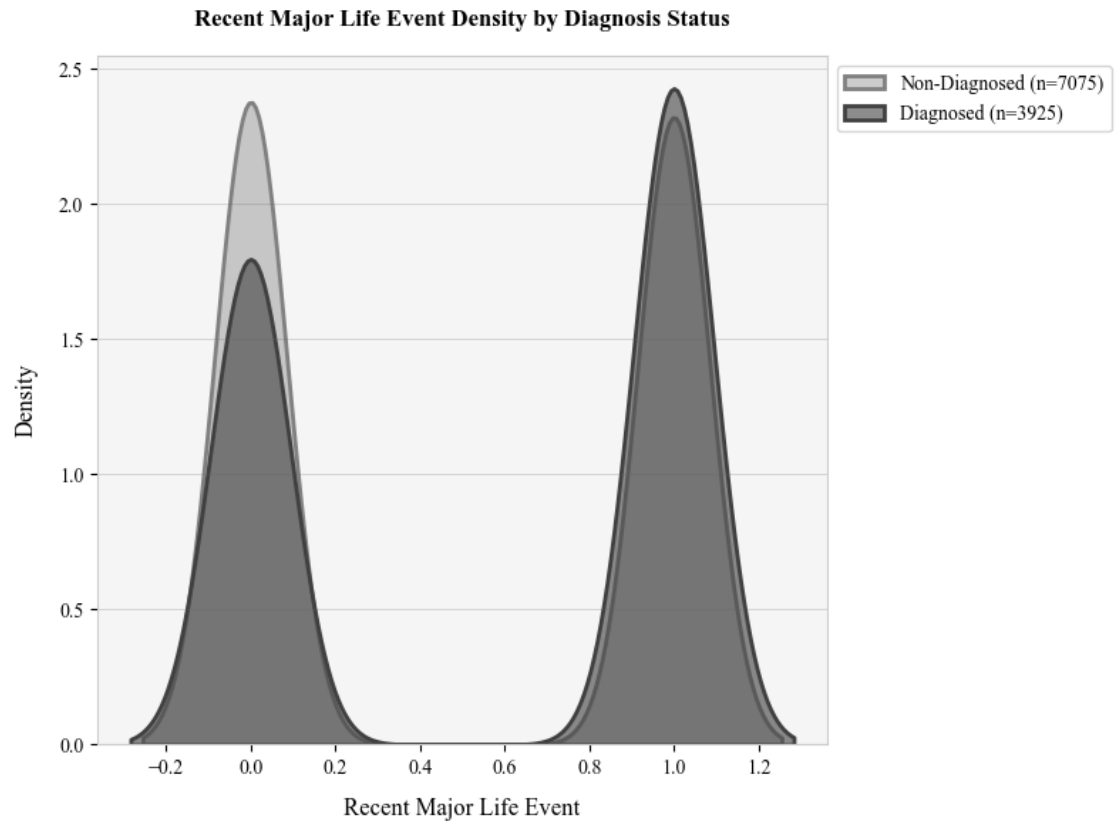


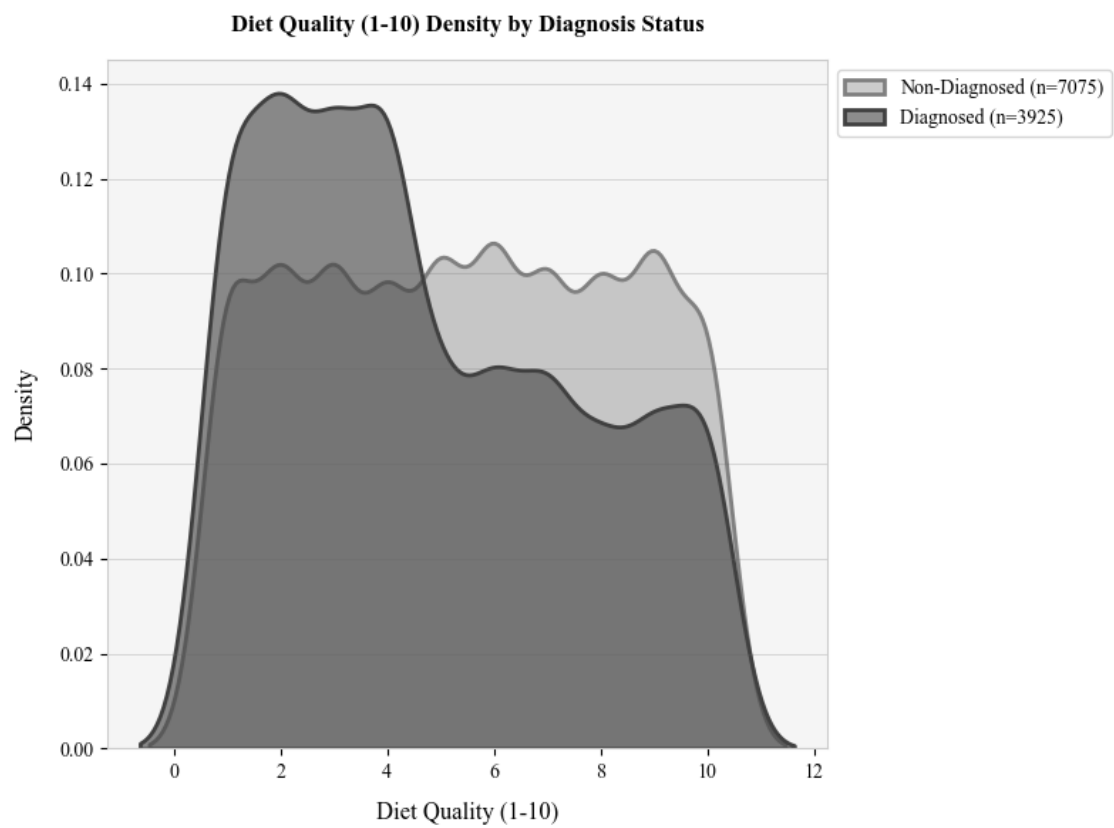


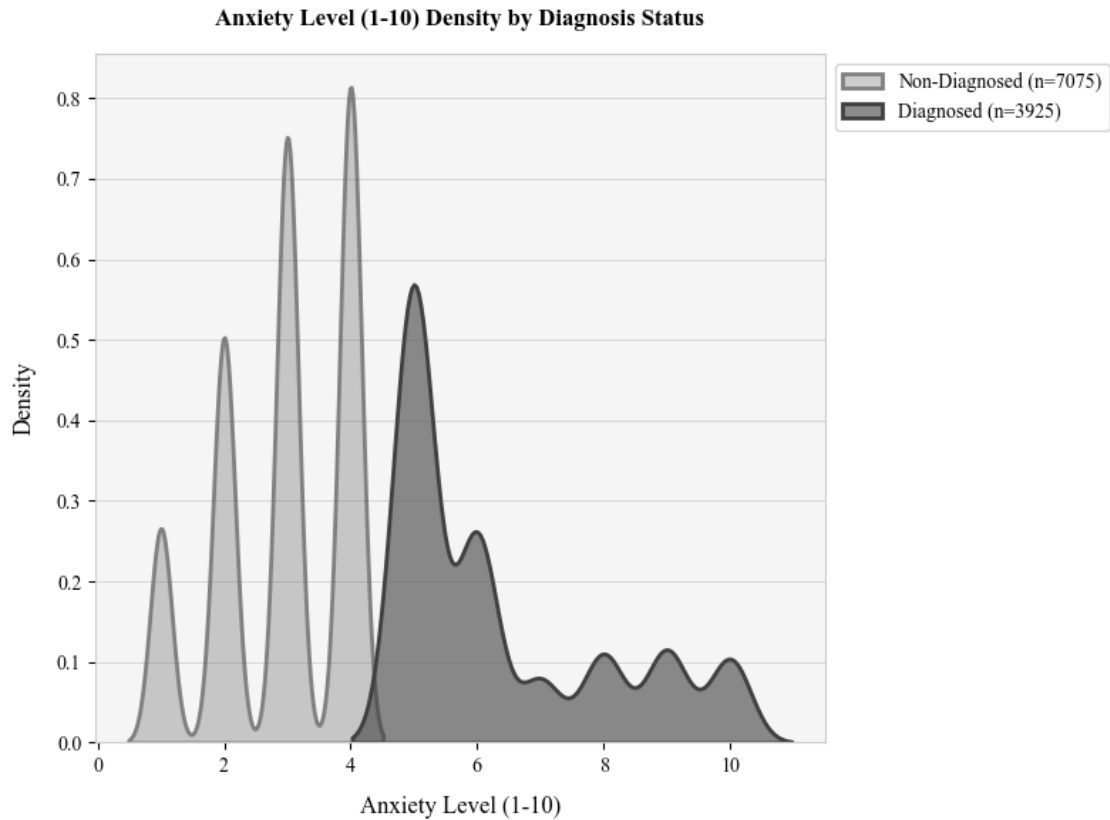












SKEWNESS AND KURTOSIS OF FEATURES

	Skewness	Kurtosis
Family History of Anxiety	2.154874	2.643963
Smoking	1.308571	-0.287695
Anxiety Level (1-10)	0.918902	0.662642
Physical Activity (hrs/week)	0.490496	-0.272559
Stress Level (1-10)	-0.148151	-1.273465
Diet Quality (1-10)	0.139806	-1.217157
Heart Rate (bpm)	-0.128128	-1.157126
Caffeine Intake (mg/day)	-0.115152	-1.186871
Therapy Sessions (per month)	-0.099686	-1.192781
Recent Major Life Event	-0.091380	-1.992012
Breathing Rate (breaths/min)	-0.090613	-1.199353
Sweating Level (1-5)	-0.088569	-1.269781
Dizziness	-0.071692	-1.995223
Age	0.061239	-1.156933
Alcohol Consumption (drinks/week)	-0.040870	-1.176549
Medication	-0.035283	-1.999119
Sleep Hours	-0.011845	-0.262601

CATEGORICAL FEATURES SUMMARY

Categorical Feature: Gender

```

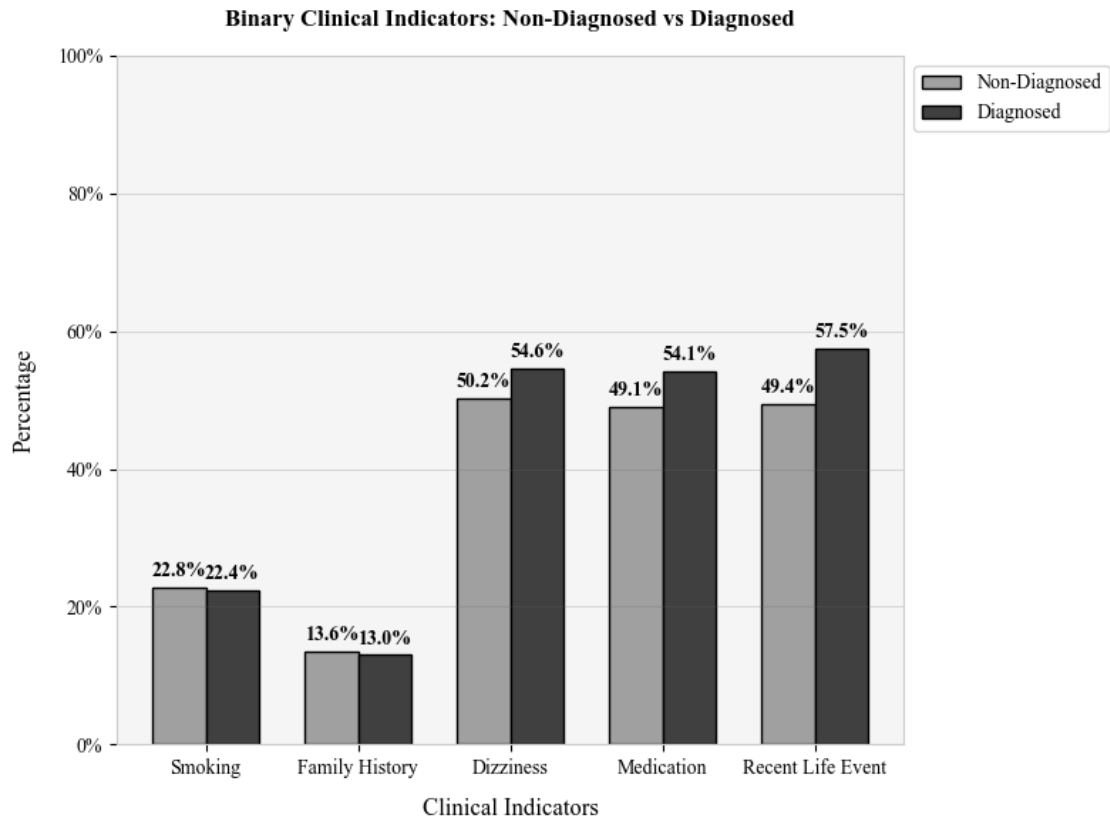
Number of unique values: 3
Value counts top 10:
Gender
Other      3706
Male       3666
Female     3628
Name: count, dtype: int64
Categorical Feature: Occupation
Number of unique values: 13
Value counts top 10:
Occupation
Scientist   880
Teacher    871
Artist     870
Engineer   868
Freelancer 864
Other      858
Lawyer     854
Athlete    847
Nurse      833
Student    826
Name: count, dtype: int64

```

```

=====
BINARY FEATURES PERCENTAGES BY DIAGNOSIS GROUP

```



Binary Features - Percentage by Diagnosis Group:

```

-----
Smoking          Non-Diagnosed: 22.8% | Diagnosed: 22.4% | Difference:
-0.4%
Family History   Non-Diagnosed: 13.6% | Diagnosed: 13.0% | Difference:
-0.6%
Dizziness        Non-Diagnosed: 50.2% | Diagnosed: 54.6% | Difference:
4.4%
Medication       Non-Diagnosed: 49.1% | Diagnosed: 54.1% | Difference:
5.0%
Recent Life Event Non-Diagnosed: 49.4% | Diagnosed: 57.5% | Difference:
8.1%

```

```

=====
Dataset saved
Final dataset shape: 11000 rows, 20 columns

```